

# Density-dependence seedling mortality in Kadumane

Robert Bagchi and Ashwin Viswanathan

13 January 2025

## Contents

|                                                 |           |
|-------------------------------------------------|-----------|
| 0.1 set up . . . . .                            | 1         |
| <b>1 Data</b>                                   | <b>1</b>  |
| <b>2 Summary information for the paper</b>      | <b>5</b>  |
| <b>3 Models</b>                                 | <b>6</b>  |
| 3.1 Scaled conspecific density models . . . . . | 6         |
| 3.2 Species specific inferences. . . . .        | 34        |
| <b>4 Species abundances</b>                     | <b>59</b> |
| <b>5 Session Information</b>                    | <b>63</b> |

## 0.1 set up

```
library(tidyverse)
library(ggthemes)
library(knitr)
library(glmmTMB)
library(DHARMA)
library(broom.mixed)
library(ggeffects)
library(ggdist)
library(sjPlot)
library(patchwork)
```

```
theme_set(theme_tufte())
```

## 1 Data

Load data

```
sp_codes <- read_csv("data/sp_codes.csv")
```

```
## Rows: 26 Columns: 4
## -- Column specification -----
## Delimiter: ","
## chr (4): code, genus, species, family
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
site_dat <- read_rds("data/kadumane_location_metadata.rds")
sdls <- read_rds("data/kadumane_seedlings.rds")
plot_dat <- read_rds("data/kadumane_plot_metadata.rds")
```

1. Join in site and location meta-data.
2. Remove rows for species with no seedlings at start of the census in plot.
3. Select species that
  - Are recorded in at least 5 plots.
  - Vary in density among plots (i.e., don't have the same variation everywhere).
  - Are groups of many (unidentified) species with the same code.

```
sdls |> summarise(n_sdls = sum(census.start),
                 n_survs = sum(census.final),
                 n_species = n_distinct(species)) |> knitr::kable()
```

| n_sdls | n_survs | n_species |
|--------|---------|-----------|
| 7134   | 3054    | 102       |

```
sdls <- sdls |> left_join(site_dat) |> left_join(plot_dat) |>
  filter(census.start > 0)
```

```
## Joining with 'by = join_by(site, location)'
## Joining with 'by = join_by(site, location, group, plot)'
```

```
## find species with enough individuals (>4) and variation in density (var > 0)
## to allow analysis
sp_list <- group_by(sdls, species) |>
  summarise(abund = sum(census.start), ## total abund
            n = sum(census.start > 0), ## number of plots with the species
            sd_dens = sd(census.start[census.start > 0])) |> ##var in density
  filter(n > 1) |> arrange(n)
## only lose 8 species by restricting to 5 or more occurrences (instead of 1)
## Seems reasonable. Also removing species that were unreliably identified.

sp_list <- filter(sp_list, n > 4, !(species %in% c("Palm", "Artoh", "SC")),
                 sd_dens > 0)
sdls <- filter(sdls, species %in% sp_list$species)
dim(sdls) ## 968 columns
```

```
## [1] 968 18
```

```
# Number of plots= 37 locs * 3 groups * 5 plots
n_plots <- 37*3*5
n_sites <- 21
```

Rename columns, add total seedling density, scale and centre data.

```
## shorten names
sdls <- rename(sdls,
               "trt_F" = "treatment.fungicide",
               "trt_I" = "treatment.insecticide",
               "Pr_m" = "proportion.mortality",
               "gr" = "group", ## group causes problems with some helper funcs
               "loc" = "location")
## categorical variable for treatment
sdls <- mutate(sdls, trt = case_when(
  trt_F == "0" & trt_I == "0" ~ "C",
  trt_F == "F" & trt_I == "0" ~ "F",
  trt_F == "0" & trt_I == "I" ~ "I",
  trt_F == "F" & trt_I == "I" ~ "FI"),
  trt = factor(trt, levels = c("C", "I", "F", "FI")))
## add total density
tot_dens <- sdls |> group_by(site, loc, gr, plot) |>
  summarise(tot_dens = sum(census.start))
```

## 'summarise()' has grouped output by 'site', 'loc', 'gr'. You can override using the '.groups' argument.

```
## add species mean density
## divide by total number of plots
sp_mean_dens <- sdls |> group_by(species) |>
  summarise(sp_mean_condens = sum(census.start)/n_plots,
            sp_mean_surv = sum(census.final)/sum(census.start))

summary(sp_mean_dens)
```

```
##      species  sp_mean_condens  sp_mean_surv
## Acacia   : 1  Min.    :0.01261  Min.    :0.1429
## Beilsch  : 1  1st Qu.:0.03559  1st Qu.:0.4396
## Cinam    : 1  Median :0.08739  Median :0.5722
## Climber  : 1  Mean    :0.43022  Mean    :0.5701
## Climber1 : 1  3rd Qu.:0.17207  3rd Qu.:0.7086
## Crypto   : 1  Max.    :6.48288  Max.    :1.0000
## (Other)  :20
```

```
sdls <- left_join(sdls, tot_dens, by = c("site", "loc", "gr", "plot")) |>
  left_join(select(sp_mean_dens, - sp_mean_surv))
```

## Joining with 'by = join\_by(species)'

```

#scale density by mean, fix couple of NAs
sdls <- mutate(sdls,
  slope.degrees = replace_na(slope.degrees, 5),
  Pr_s = 1 - Pr_m,
  con_dens = census.start,
  con_dens_s = con_dens/sp_mean_condens,
  slope.degrees_s = as.vector(scale(slope.degrees)),
  trt_F = factor(trt_F, labels = c("O", "F")),
  trt_I = factor(trt_I, labels = c("O", "I"))
)

## calculate total scaled density. Note this is the sum of the scaled densities
## of all species in the plot. Remember that the total_density contrast only
## works when all conspecific densities sum to total density (replicating
## a sum-to-zero contrast for conspecific and heterospecific densities.
sdls <- left_join(sdls,
  group_by(sdls, site, loc, gr, plot) |>
  summarise(tot_dens_s = sum(con_dens_s)))

## 'summarise()' has grouped output by 'site', 'loc', 'gr'. You can override using
## the '.groups' argument.
## Joining with 'by = join_by(site, loc, gr, plot)'

```

```
summary(sdls)
```

```

##      site           loc           gr      plot      species
## Length:968      Length:968      G1:283    1:185    SR       :172
## Class :character Class :character G2:351    2:210    Symp      :122
## Mode  :character Mode  :character G3:334    5:183    Cinam     : 86
##                                     6:199    Climber1: 77
##                                     7:191    FLC       : 64
##                                     Litsea    : 61
##                                     (Other)   :386
## census.start      census.mid      census.final      Pr_m
## Min.   : 1.000    Min.   : 0.000    Min.   : 0.000    Min.   :0.0000
## 1st Qu.: 1.000    1st Qu.: 0.000    1st Qu.: 0.000    1st Qu.:0.0000
## Median : 1.000    Median : 1.000    Median : 1.000    Median :0.1667
## Mean   : 6.413    Mean   : 3.089    Mean   : 2.769    Mean   :0.3810
## 3rd Qu.: 3.000    3rd Qu.: 2.000    3rd Qu.: 2.000    3rd Qu.:1.0000
## Max.   :300.000    Max.   :109.000    Max.   :90.000    Max.   :1.0000
##
## fragment.size     edge.distance   location.code   group.code   proportion.rock
## Min.   : 1.10     Min.   : 50.0    S14L1 : 54     S14L1G2: 23    Min.   :0.0000
## 1st Qu.: 9.00     1st Qu.: 60.0    S1L2  : 49     S12L1G2: 18    1st Qu.:0.0500
## Median : 46.00     Median : 90.0    S12L2 : 40     S14L1G3: 17    Median :0.1000
## Mean   : 51.15     Mean   :109.3    S13L3 : 40     S1L1G3 : 17    Mean   :0.1888
## 3rd Qu.: 64.50     3rd Qu.:150.0    S12L1 : 37     S1L2G1 : 17    3rd Qu.:0.3000
## Max.   :149.00     Max.   :260.0    S3L3  : 37     S1L2G2 : 17    Max.   :0.9000
##                                     (Other):711    (Other):859    NA's   :3
## slope.degrees      treatment.control trt_F   trt_I   trt      tot_dens
## Min.   : 0.000     Min.   :0.0000    0:594  0:578  C       :395    Min.   : 1.00
## 1st Qu.: 5.000     1st Qu.:0.0000    F:374  I:390  I       : 0     1st Qu.: 3.00
## Median : 5.000     Median :0.0000                      F       : 0     Median : 6.00

```

```
## Mean : 9.199 Mean :0.4081 FI : 0 Mean : 15.79
## 3rd Qu.:10.000 3rd Qu.:1.0000 NA's:573 3rd Qu.: 14.00
## Max. :40.000 Max. :1.0000 Max. :301.00
##
## sp_mean_condens Pr_s con_dens con_dens_s
## Min. :0.01261 Min. :0.0000 Min. : 1.000 Min. : 0.1543
## 1st Qu.:0.12793 1st Qu.:0.0000 1st Qu.: 1.000 1st Qu.: 1.7732
## Median :0.24685 Median :0.8333 Median : 1.000 Median : 6.8841
## Mean :1.49738 Mean :0.6190 Mean : 6.413 Mean : 14.9070
## 3rd Qu.:1.14595 3rd Qu.:1.0000 3rd Qu.: 3.000 3rd Qu.: 13.2143
## Max. :6.48288 Max. :1.0000 Max. :300.000 Max. :355.2000
##
## slope.degrees_s tot_dens_s
## Min. :-1.08744 Min. : 0.1543
## 1st Qu.: -0.49640 1st Qu.: 9.8715
## Median : -0.49640 Median : 23.5926
## Mean : 0.00000 Mean : 39.0650
## 3rd Qu.: 0.09464 3rd Qu.: 47.7227
## Max. : 3.64087 Max. :357.8179
##
```

```
dim(sdls) ## 968 species x plot combinations
```

```
## [1] 968 26
```

## 2 Summary information for the paper

```
sdls |> summarise(n_sdls = sum(census.start),
  n_survs = sum(census.final),
  n_species = n_distinct(species)) |> knitr::kable()
```

| n_sdls | n_survs | n_species |
|--------|---------|-----------|
| 6208   | 2680    | 26        |

```
sdls <- droplevels(sdls)
table(sdls$species)[order(table(sdls$species))]
```

```
##
## Here SN Humb Crypto Holi Mac Rattan Toona
## 5 5 6 7 7 8 8 8
## SP Beilsch Acacia Mallotus Mel Trich Poly UID20
## 9 12 15 16 19 21 26 31
## Olea Climber Psyflav Dimo Litsea FLC Climber1 Cinam
## 36 45 46 56 61 64 77 86
## Symp SR
## 122 172
```

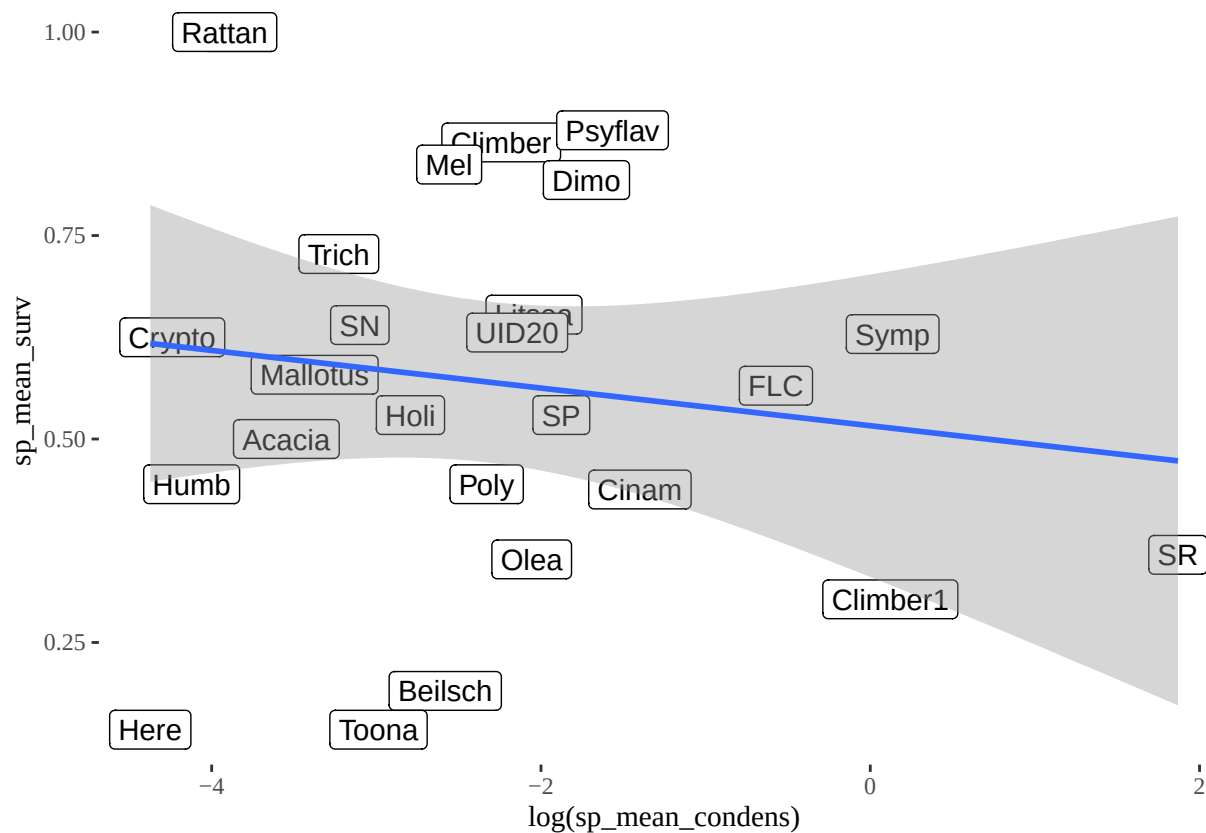
## 3 Models

### 3.1 Scaled conspecific density models

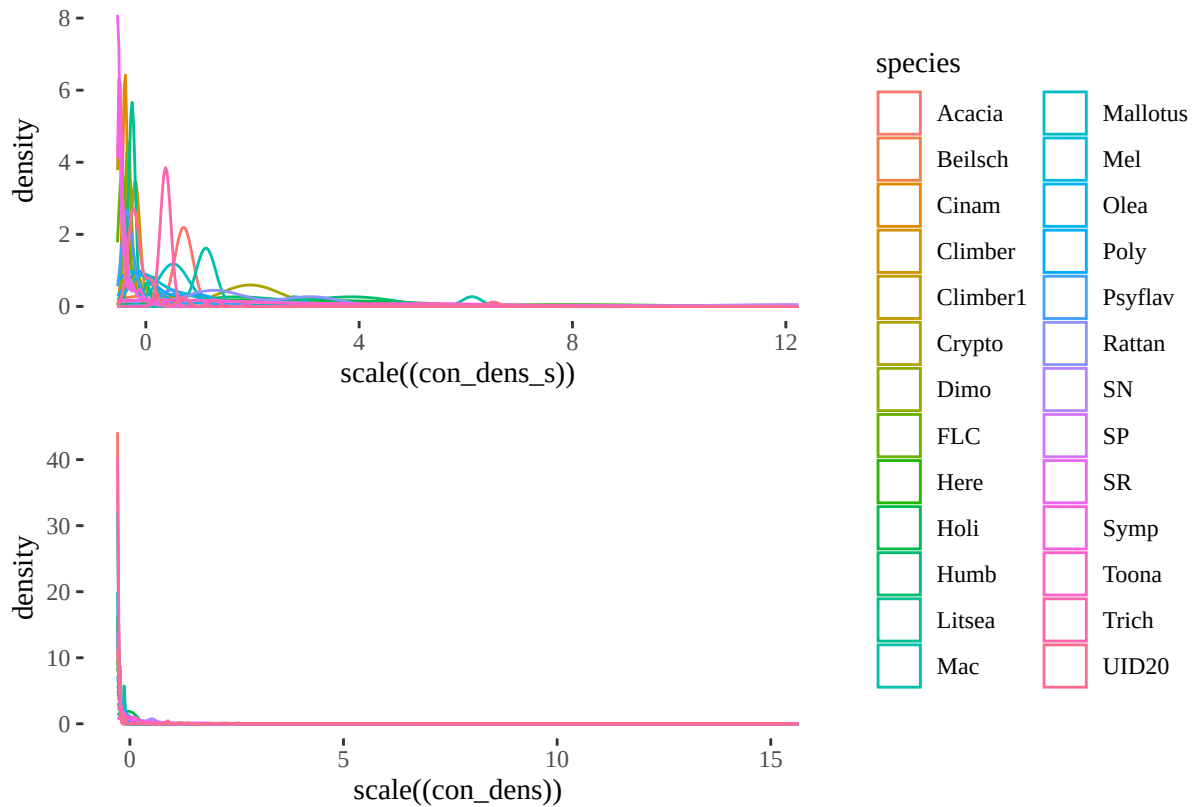
The most abundant species initially will often have lower survival (fecundity/ survival trade-off). This could generate what looks like a density-dependent relationship when looking across species, even without a relationship within species (i.e., Simpson's paradox).

```
ggplot(sp_mean_dens, aes(x = log(sp_mean_condens), y = sp_mean_surv)) +  
  geom_label(aes(label= species)) + geom_smooth(method = "lm") + theme_tufte()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## not the clearest pattern, but worth accounting for, especially given that  
## SR is so influential.  
(ggplot(sdlis, aes(x = scale((con_dens_s)), colour = species)) + geom_density())/   
(ggplot(sdlis, aes(x = scale((con_dens)), colour = species)) +  
  geom_density()) + plot_layout(guides = "collect")
```



To account for this, we can scale conspecific density by dividing by mean density and refitting the models.

```
## Random intercept model
m_cdd_s_ri <- glmmTMB(Pr_s ~ slope.degrees_s +
  trt_I:trt_F +
  (scale(tot_dens_s) + scale(con_dens_s)) *
  (trt_I + trt_F) *
  scale(fragment.size) +
  (1|species) +
  (1|site/loc/gr/plot),
  weights = census.start, data = sdfs,
  family=binomial)

## Random intercept and slope model for species specific effects
m_cdd_s_ris <- glmmTMB(Pr_s ~ slope.degrees_s +
  trt_I:trt_F +
  (scale(tot_dens_s) + scale(con_dens_s)) *
  (trt_I + trt_F) *
  scale(fragment.size) +
  (scale(con_dens_s) + scale(tot_dens_s)|species) +
  (1|site/loc/gr/plot),
  weights = census.start,
  data = sdfs,
  family=binomial)

## note the slightly odd ordering of terms doesn't change the model structure,
## but does change the default ordering of terms in outputs and plots to a
```

```
## more convenient one for describing in paper (first CDD, then biocide effects
## then fragmentation effects).
```

```
anova(m_cdd_s_ri, m_cdd_s_ris) ## random slope *much* better
```

```
## Data: sdls
## Models:
## m_cdd_s_ri: Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) * , zi=~0
## m_cdd_s_ri:      (trt_I + trt_F) * scale(fragment.size) + (1 | species) + , zi=~0, disp=~1
## m_cdd_s_ri:      (1 | site/loc/gr/plot), zi=~0, disp=~1
## m_cdd_s_ris: Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) * , zi=~0
## m_cdd_s_ris:      (trt_I + trt_F) * scale(fragment.size) + (scale(con_dens_s) + , zi=~0, disp=~1
## m_cdd_s_ris:      scale(tot_dens_s) | species) + (1 | site/loc/gr/plot), zi=~0, disp=~1
##           Df      AIC      BIC logLik deviance Chisq Chi Df Pr(>Chisq)
## m_cdd_s_ri  25 2232.1 2353.9 -1091.0   2182.1
## m_cdd_s_ris  30 2215.1 2361.3 -1077.5   2155.1 27.02      5 5.654e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Models fit without issues, even with the correlation among density effects.

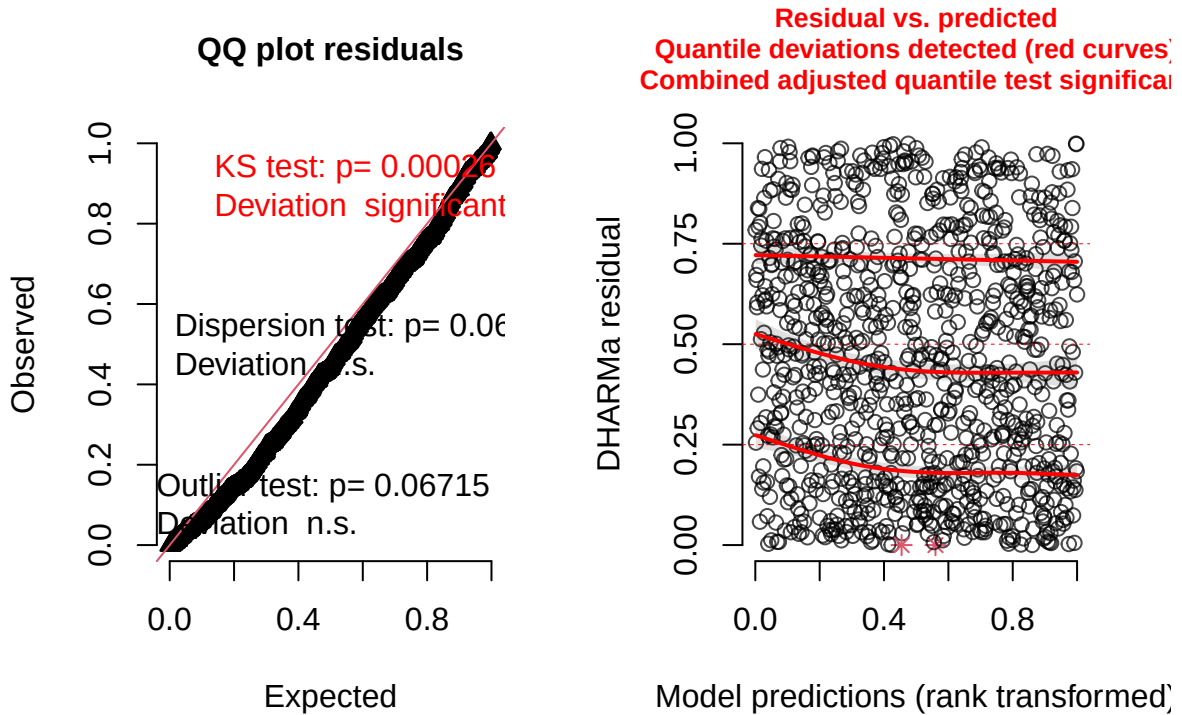
The improvement with the random slopes model suggests we need to look at individual species. Proceeding with the random intercept and slope model from here on.

### 3.1.1 Diagnostics

```
res_s <- simulateResiduals(m_cdd_s_ris)
plot(res_s) ## ok - some deviation from ideal residual distribution, but
```



## DHARMA residual



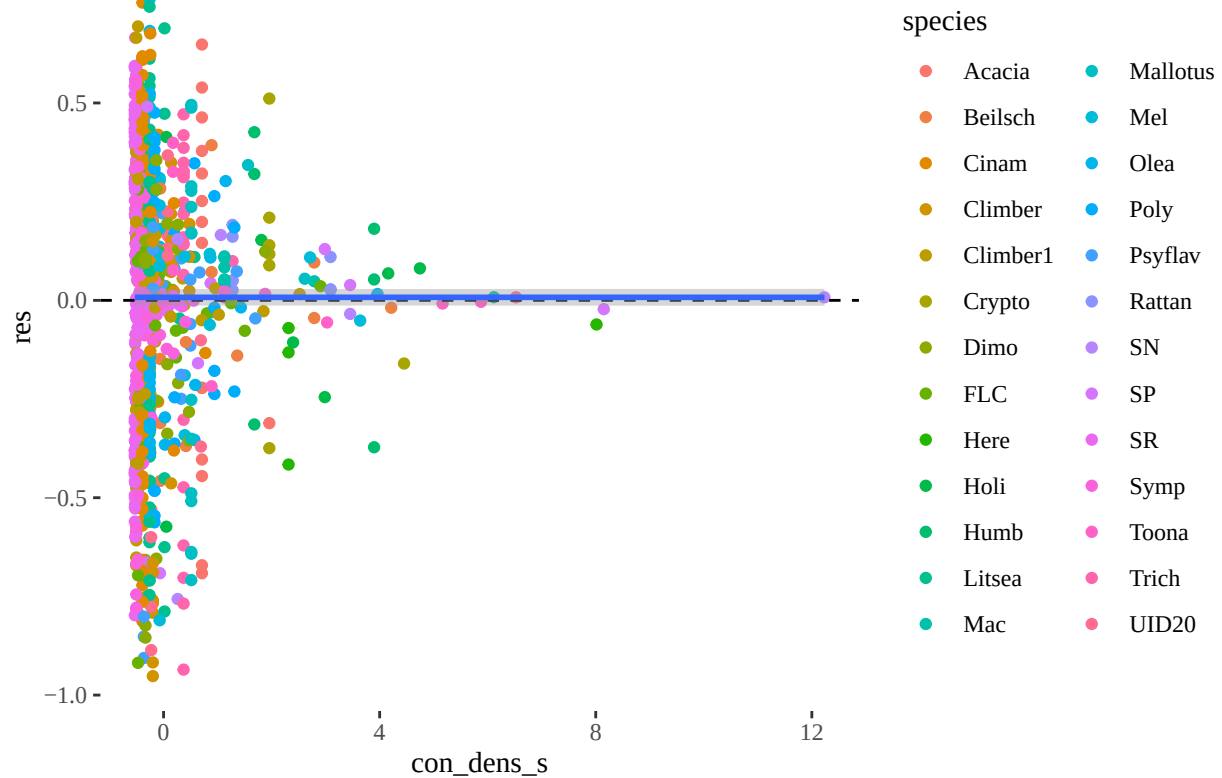
```
## acceptable.

## look at relationship with covariates
diag_dat <- data.frame(m_cdd_s_ris$frame, res = res_s$fittedResiduals)

diag_dat <- rename_with(diag_dat, ~ str_replace(.x, "scale\\.", "")) |>
  rename_with(~str_replace(.x, "\\.$", ""))

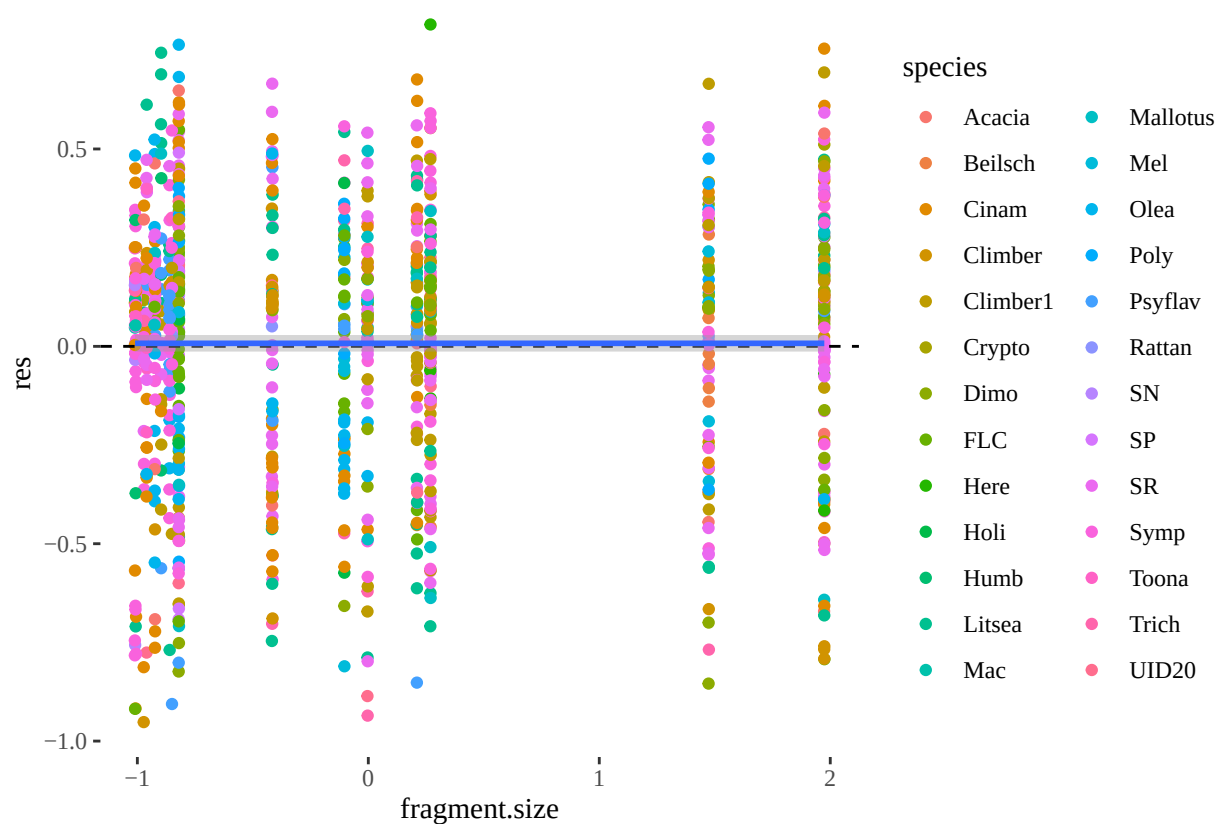
ggplot(diag_dat, aes(x = con_dens_s, y = res)) +
  geom_point(aes(colour = species)) +
  geom_hline(yintercept=0, linetype = "dashed") +
  geom_smooth(method="gam") ## no trend.

## 'geom_smooth()' using formula = 'y ~ s(x, bs = "cs")'
```



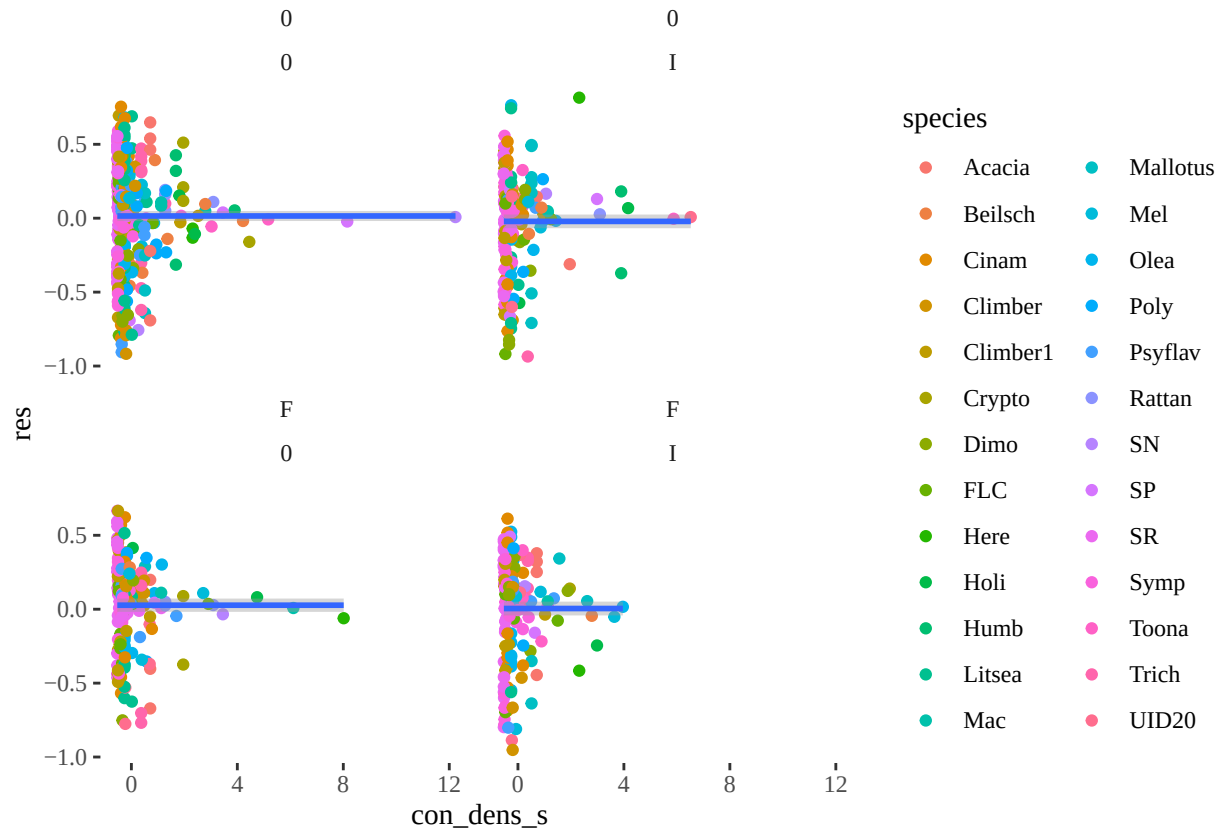
```
ggplot(diag_dat, aes(x = fragment.size, y = res)) +
  geom_point(aes(colour = species), position = "jitter") +
  geom_hline(yintercept=0, linetype = "dashed") +
  geom_smooth(method="gam") ## no trend.
```

```
## 'geom_smooth()' using formula = 'y ~ s(x, bs = "cs")'
```



```
ggplot(diag_dat, aes(x = con_dens_s, y = res)) +
  facet_wrap(~trt_F + trt_I ) +
  geom_point(aes(colour = species)) +
  geom_smooth(method="gam") ## no trend with treatment
```

```
## 'geom_smooth()' using formula = 'y ~ s(x, bs = "cs")'
```



The diagnostics aren't perfect, but not particularly unusual for a binomial model. None of the big problems (e.g., overdispersion, trends with covariates) seem to apply here.

### 3.1.2 Model inference

```
summary(m_cdd_s_ris)
```

```
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) *
## (trt_I + trt_F) * scale(fragment.size) + (scale(con_dens_s) +
## scale(tot_dens_s) | species) + (1 | site/loc/gr/plot)
## Data: sdls
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
##  2215.0   2361.3  -1077.5   2155.0     938
##
## Random effects:
##
## Conditional model:
## Groups      Name                Variance Std.Dev.  Corr
## species      (Intercept)         1.027e+00 1.0131846
##              scale(con_dens_s)  3.887e-01 0.6234621  0.56
```

```

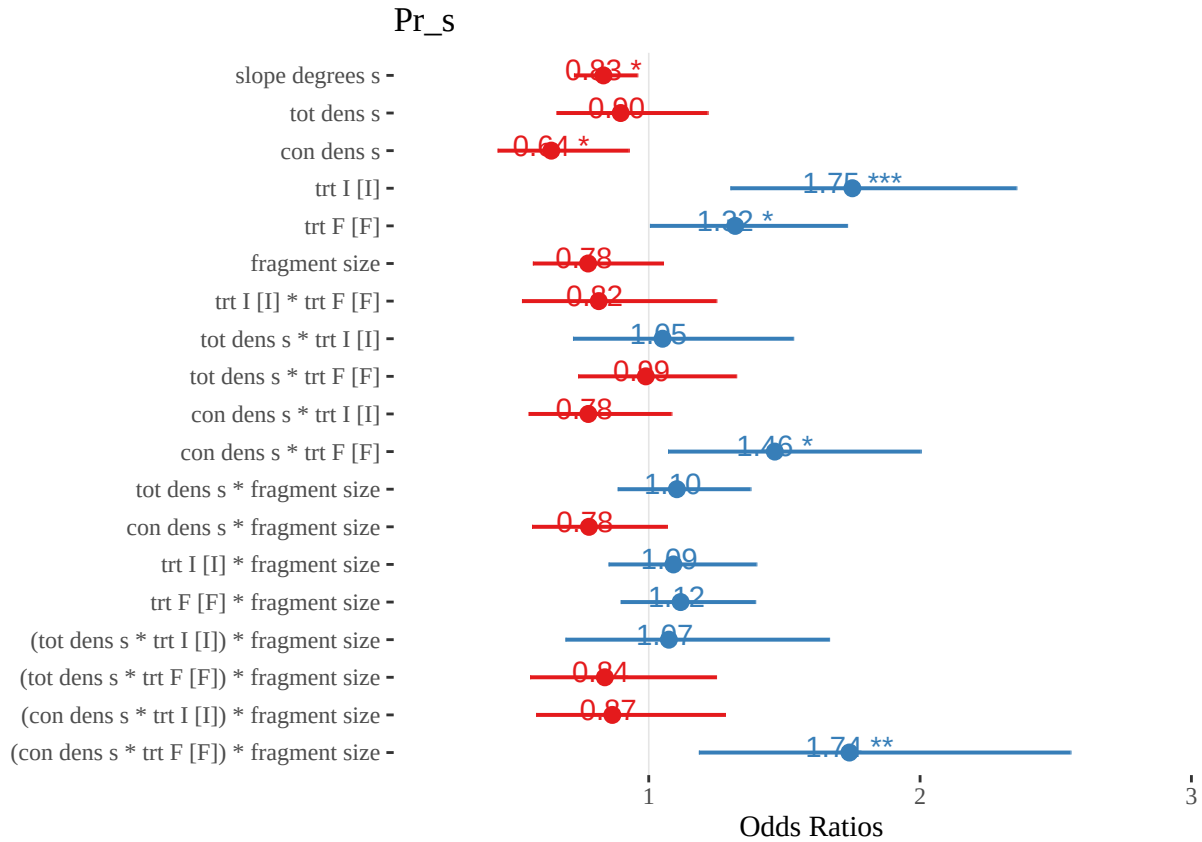
##          scale(tot_dens_s) 3.738e-02 0.1933384 -0.47 -0.71
## plot:gr:loc:site (Intercept) 2.788e-01 0.5280003
## gr:loc:site      (Intercept) 2.057e-01 0.4535170
## loc:site         (Intercept) 4.782e-01 0.6915250
## site             (Intercept) 2.188e-08 0.0001479
## Number of obs: 968, groups:
## species, 26; plot:gr:loc:site, 474; gr:loc:site, 110; loc:site, 37; site, 21
##
## Conditional model:
##
##          Estimate Std. Error z value
## (Intercept)          0.66723    0.27771    2.403
## slope.degrees_s      -0.18268    0.07133   -2.561
## scale(tot_dens_s)     -0.10921    0.15595   -0.700
## scale(con_dens_s)     -0.44496    0.18820   -2.364
## trt_II                0.55988    0.15137    3.699
## trt_FF                0.27661    0.13886    1.992
## scale(fragment.size)  -0.25279    0.15514   -1.629
## trt_II:trt_FF         -0.20355    0.21738   -0.936
## scale(tot_dens_s):trt_II 0.04971    0.19245    0.258
## scale(tot_dens_s):trt_FF -0.01093    0.14816   -0.074
## scale(con_dens_s):trt_II -0.25189    0.16952   -1.486
## scale(con_dens_s):trt_FF 0.38152    0.15970    2.389
## scale(tot_dens_s):scale(fragment.size) 0.09861    0.11221    0.879
## scale(con_dens_s):scale(fragment.size) -0.24883    0.16012   -1.554
## trt_II:scale(fragment.size) 0.08701    0.12608    0.690
## trt_FF:scale(fragment.size) 0.11108    0.11225    0.990
## scale(tot_dens_s):trt_II:scale(fragment.size) 0.07160    0.22351    0.320
## scale(tot_dens_s):trt_FF:scale(fragment.size) -0.17654    0.20314   -0.869
## scale(con_dens_s):trt_II:scale(fragment.size) -0.14412    0.19987   -0.721
## scale(con_dens_s):trt_FF:scale(fragment.size) 0.55367    0.19600    2.825
##
##          Pr(>|z|)
## (Intercept)          0.016277 *
## slope.degrees_s      0.010437 *
## scale(tot_dens_s)     0.483771
## scale(con_dens_s)     0.018061 *
## trt_II                0.000217 ***
## trt_FF                0.046363 *
## scale(fragment.size)  0.103231
## trt_II:trt_FF         0.349094
## scale(tot_dens_s):trt_II 0.796159
## scale(tot_dens_s):trt_FF 0.941199
## scale(con_dens_s):trt_II 0.137312
## scale(con_dens_s):trt_FF 0.016894 *
## scale(tot_dens_s):scale(fragment.size) 0.379537
## scale(con_dens_s):scale(fragment.size) 0.120172
## trt_II:scale(fragment.size) 0.490132
## trt_FF:scale(fragment.size) 0.322352
## scale(tot_dens_s):trt_II:scale(fragment.size) 0.748722
## scale(tot_dens_s):trt_FF:scale(fragment.size) 0.384811
## scale(con_dens_s):trt_II:scale(fragment.size) 0.470860
## scale(con_dens_s):trt_FF:scale(fragment.size) 0.004731 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
plot_model(m_cdd_s_ris, show.values=TRUE) + ylim(c(0.2, 3))
```

## Scale for y is already present.

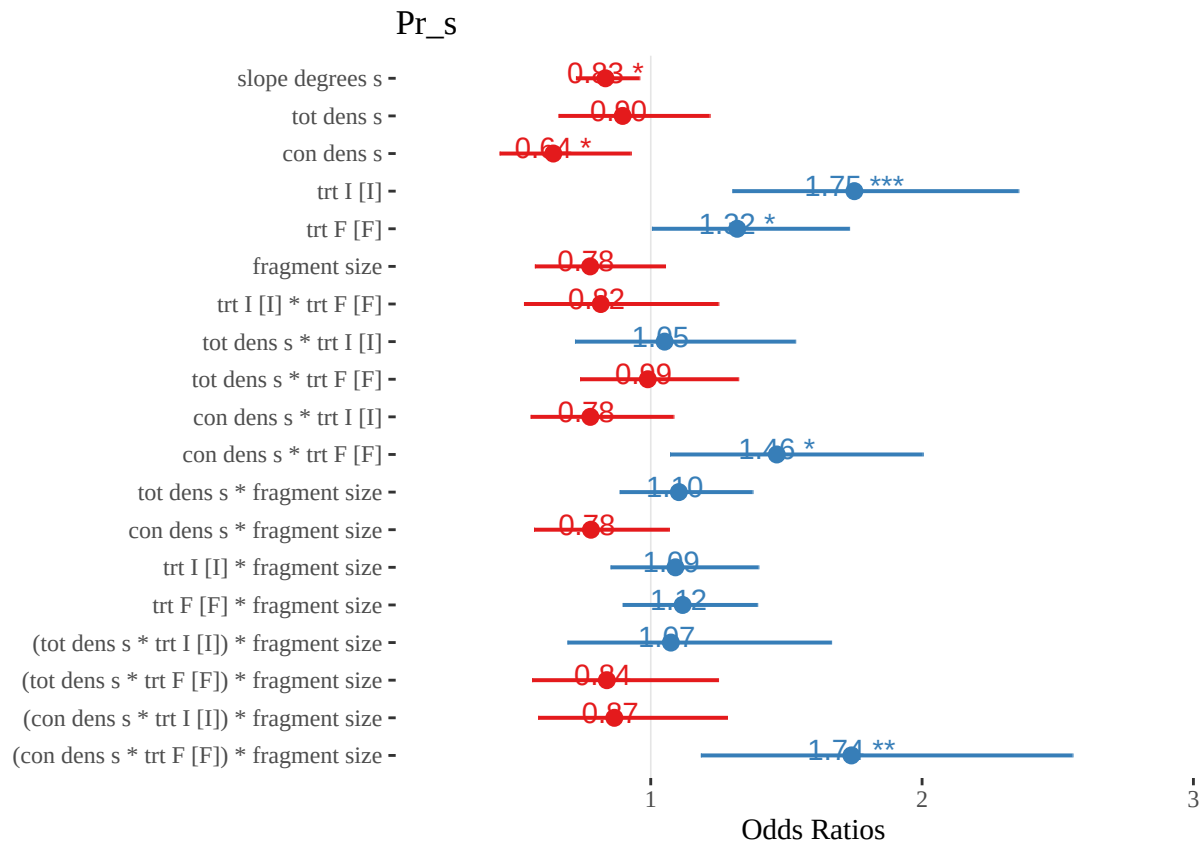
## Adding another scale for y, which will replace the existing scale.



```
plot_model(m_cdd_s_ris, show.values=TRUE) + ylim(c(0.2, 3))
```

## Scale for y is already present.

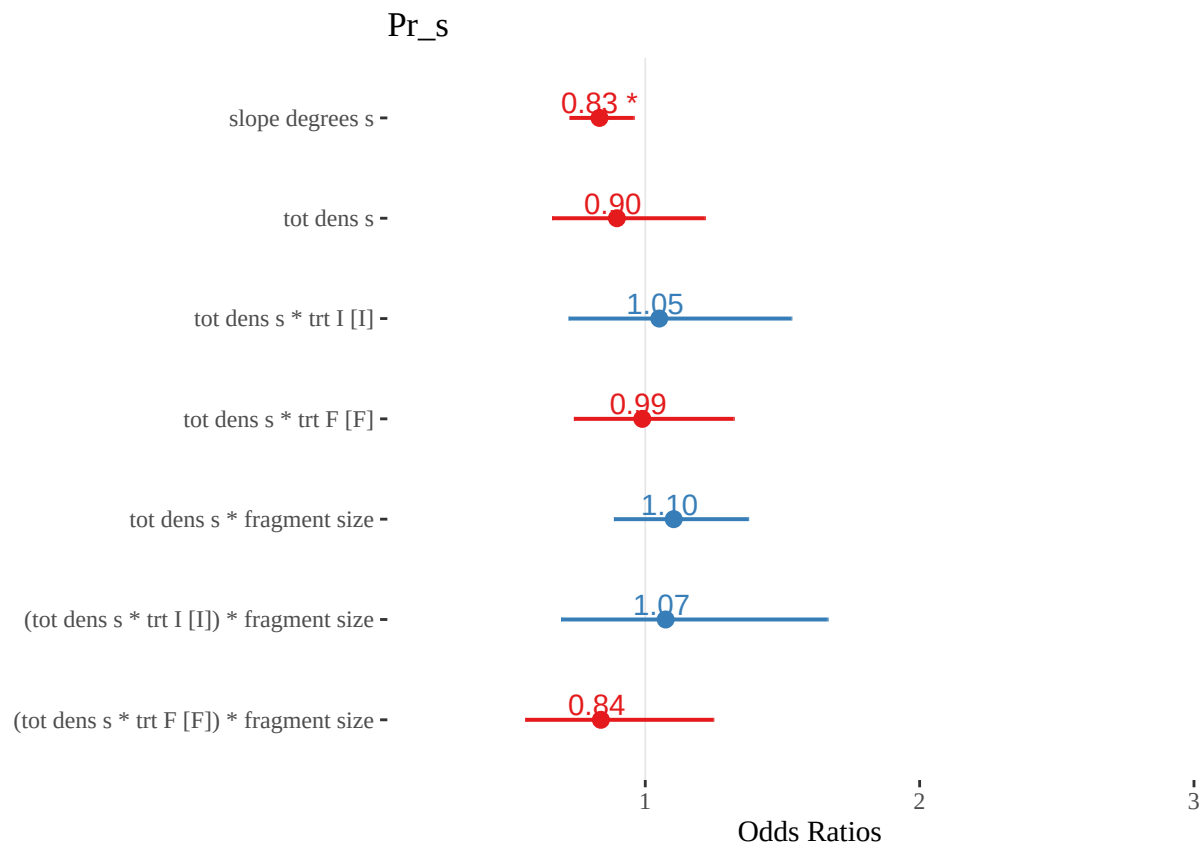
## Adding another scale for y, which will replace the existing scale.



```
## total density is never very important, so separating out it's effects
rmvars <- names(fixef(m_cdd_s_ris)$cond)
rmvars <- c(rmvars[grepl("tot_dens_s", rmvars)], "slope.degrees_s")

## First confirm it isn't influential
plot_model(m_cdd_s_ris, show.values=TRUE, terms=rmvars) + ylim(c(0.2, 3))
```

```
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.
```



```
# basically what we thought
```

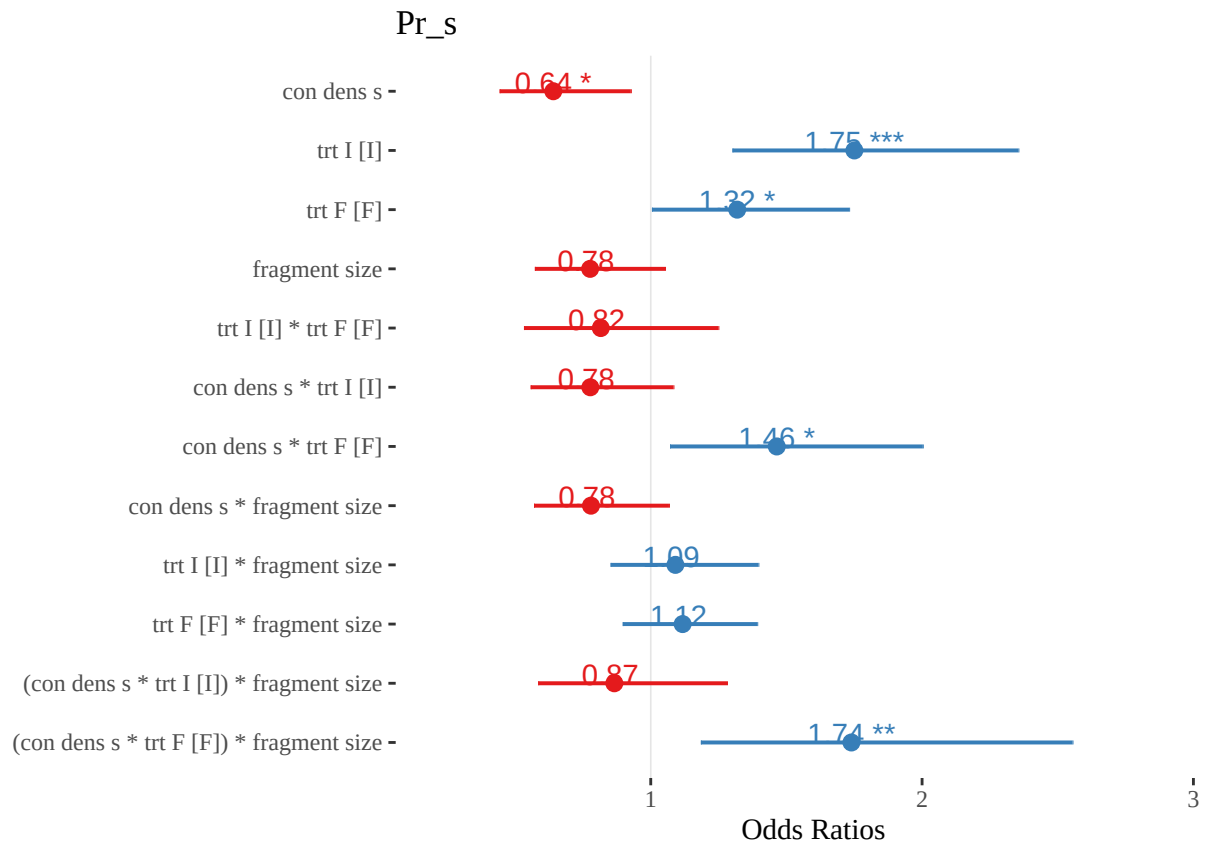
```
## now decluttered version
```

```
plot_model(m_cdd_s_ris, show.values=TRUE, rm.terms=rmvars) + ylim(c(0.2, 3))
```

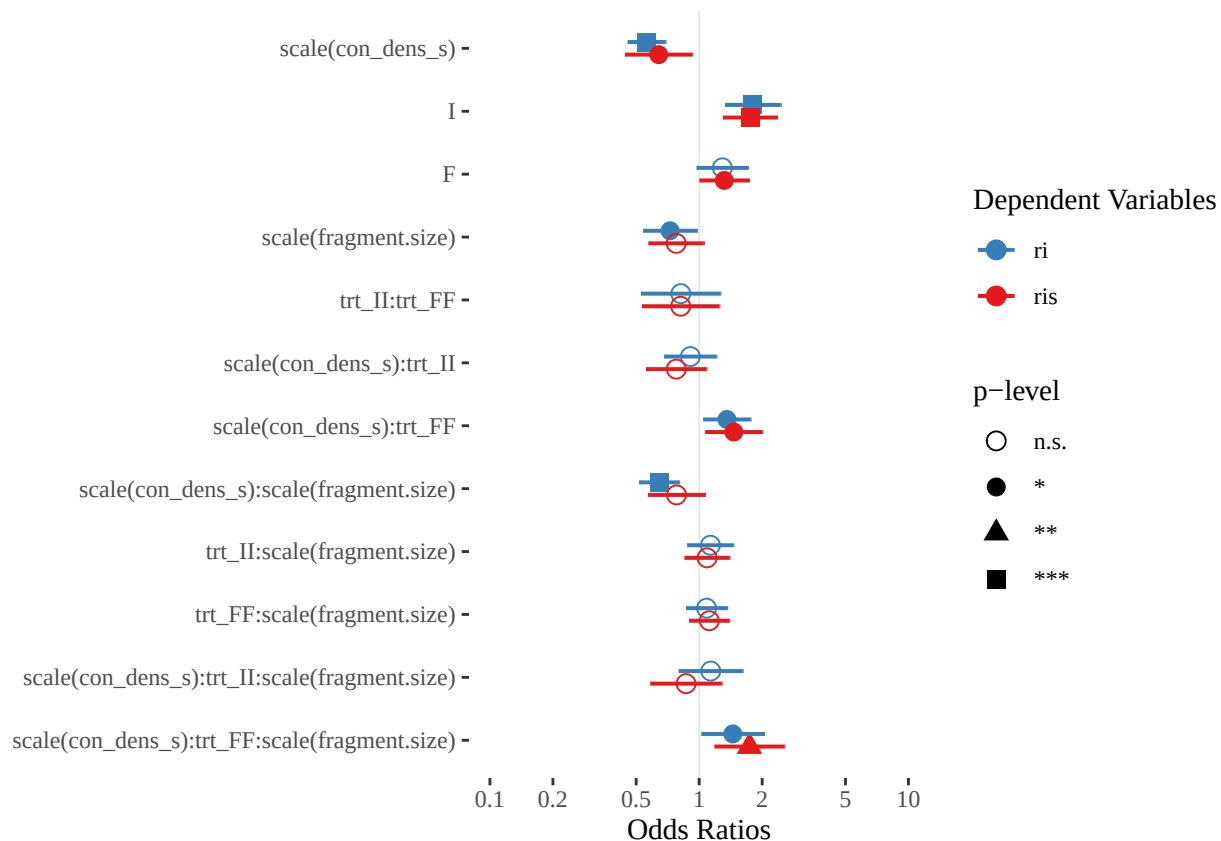
```
## Scale for y is already present.
```

```
## Adding another scale for y, which will replace the existing scale.
```





```
## compare to random intercept model
plot_models(m_cdd_s_ri, m_cdd_s_ris, rm.terms=rmvars, m.labels=c("ri", "ris"),
            p.shape = TRUE)
```



```
summary(m_cdd_s_ri)
```

```
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) *
##      (trt_I + trt_F) * scale(fragment.size) + (1 | species) +
##      (1 | site/loc/gr/plot)
## Data: sdls
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
##  2232.1   2353.9  -1091.0   2182.1     943
##
## Random effects:
##
## Conditional model:
## Groups      Name          Variance Std.Dev.
## species      (Intercept)  1.496e+00 1.2230929
## plot:gr:loc:site (Intercept) 3.554e-01 0.5961589
## gr:loc:site    (Intercept) 2.921e-01 0.5404754
## loc:site       (Intercept) 4.210e-01 0.6488833
## site           (Intercept) 1.522e-08 0.0001234
## Number of obs: 968, groups:
## species, 26; plot:gr:loc:site, 474; gr:loc:site, 110; loc:site, 37; site, 21
##
```

```

## Conditional model:
##
## Estimate Std. Error z value
## (Intercept) 0.81111 0.30834 2.631
## slope.degrees_s -0.18928 0.07399 -2.558
## scale(tot_dens_s) 0.06247 0.11818 0.529
## scale(con_dens_s) -0.57768 0.10619 -5.440
## trt_II 0.59141 0.15620 3.786
## trt_FF 0.25563 0.14449 1.769
## scale(fragment.size) -0.32059 0.15085 -2.125
## trt_II:trt_FF -0.20244 0.22300 -0.908
## scale(tot_dens_s):trt_II 0.04069 0.19865 0.205
## scale(tot_dens_s):trt_FF -0.07806 0.15129 -0.516
## scale(con_dens_s):trt_II -0.09714 0.14638 -0.664
## scale(con_dens_s):trt_FF 0.30573 0.13274 2.303
## scale(tot_dens_s):scale(fragment.size) 0.04495 0.11292 0.398
## scale(con_dens_s):scale(fragment.size) -0.43936 0.11188 -3.927
## trt_II:scale(fragment.size) 0.12382 0.12955 0.956
## trt_FF:scale(fragment.size) 0.08116 0.11529 0.704
## scale(tot_dens_s):trt_II:scale(fragment.size) 0.01694 0.22765 0.074
## scale(tot_dens_s):trt_FF:scale(fragment.size) -0.07885 0.19887 -0.396
## scale(con_dens_s):trt_II:scale(fragment.size) 0.12721 0.17947 0.709
## scale(con_dens_s):trt_FF:scale(fragment.size) 0.36934 0.17565 2.103
## Pr(>|z|)
## (Intercept) 0.008524 **
## slope.degrees_s 0.010518 *
## scale(tot_dens_s) 0.597063
## scale(con_dens_s) 5.33e-08 ***
## trt_II 0.000153 ***
## trt_FF 0.076857 .
## scale(fragment.size) 0.033566 *
## trt_II:trt_FF 0.363984
## scale(tot_dens_s):trt_II 0.837712
## scale(tot_dens_s):trt_FF 0.605885
## scale(con_dens_s):trt_II 0.506951
## scale(con_dens_s):trt_FF 0.021267 *
## scale(tot_dens_s):scale(fragment.size) 0.690614
## scale(con_dens_s):scale(fragment.size) 8.60e-05 ***
## trt_II:scale(fragment.size) 0.339178
## trt_FF:scale(fragment.size) 0.481443
## scale(tot_dens_s):trt_II:scale(fragment.size) 0.940699
## scale(tot_dens_s):trt_FF:scale(fragment.size) 0.691760
## scale(con_dens_s):trt_II:scale(fragment.size) 0.478457
## scale(con_dens_s):trt_FF:scale(fragment.size) 0.035493 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

### 3.1.3 Take-homes:

- Survival is negatively conspecific density dependent. This result is consistent across all permutations of the model tried.
- Insecticide and fungicide (to a lesser extent) increase survival significantly.
- Fungicide removes the CDD, and that effect strengthens with fragment area. However, that effect

needs more exploration (see below).

- The interaction between CDD and fragment size is complex, but perhaps explained by species turnover. The CDD effect strengthens with fragment area, but not significantly (or particularly close) in the (better) random slope model. In the random intercept model, the relationship is significant. This is probably because of species turnover - perhaps species that prefer larger fragments happen to be ones that show stronger density dependence?
- Effects of fragment area and any interactions with density need more probing as the effects are sensitive to model structure.

### 3.1.4 adding the edge effect term

Reviewers requested consideration of edge effects in the model. Doing so here

```
## First of all, is there a correlation between edge effects and fragment area?
with(site_dat, cor.test(edge.distance, fragment.size)) ## mild correlation
```

```
##
## Pearson's product-moment correlation
##
## data: edge.distance and fragment.size
## t = 1.814, df = 35, p-value = 0.07826
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.03410824 0.56363220
## sample estimates:
## cor
## 0.2931506
```

```
m_cdde_s_ris <- glmmTMB(Pr_s ~ slope.degrees_s +
  trt_I:trt_F +
  (scale(tot_dens_s) + scale(con_dens_s)) *
  (trt_I + trt_F) *
  (scale(fragment.size) + scale(edge.distance)) +
  (scale(con_dens_s) + scale(tot_dens_s)|species) +
  scale(fragment.size) * scale(edge.distance) +
  (1|site/loc/gr/plot),
  weights = census.start,
  data = sdls,
  family=binomial)
anova(m_cdde_s_ris, m_cdd_s_ris) ## edge effects don't improve the model
```

```
## Data: sdls
## Models:
## m_cdd_s_ris: Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) * , zi=~0, disp=~1
## m_cdd_s_ris: (trt_I + trt_F) * scale(fragment.size) + (scale(con_dens_s) + , zi=~0, disp=~1
## m_cdd_s_ris: scale(tot_dens_s) | species) + (1 | site/loc/gr/plot), zi=~0, disp=~1
## m_cdde_s_ris: Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) * , zi=~0, disp=~1
## m_cdde_s_ris: (trt_I + trt_F) * (scale(fragment.size) + scale(edge.distance)) + , zi=~0, disp=~1
## m_cdde_s_ris: (scale(con_dens_s) + scale(tot_dens_s) | species) + scale(fragment.size) * , zi=~0, disp=~1
## m_cdde_s_ris: scale(edge.distance) + (1 | site/loc/gr/plot), zi=~0, disp=~1
## Df AIC BIC logLik deviance Chisq Chi Df Pr(>Chisq)
```

```
## m_cdd_s_ris 30 2215.1 2361.3 -1077.5 2155.1
## m_cdde_s_ris 40 2227.0 2422.0 -1073.5 2147.0 8.0135 10 0.6275
```

```
summary(m_cdde_s_ris)
```

```
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) *
## (trt_I + trt_F) * (scale(fragment.size) + scale(edge.distance)) +
## (scale(con_dens_s) + scale(tot_dens_s) | species) + scale(fragment.size) *
## scale(edge.distance) + (1 | site/loc/gr/plot)
## Data: sdls
## Weights: census.start
##
##      AIC      BIC    logLik deviance df.resid
## 2227.0    2422.0 -1073.5   2147.0      928
##
## Random effects:
##
## Conditional model:
## Groups          Name              Variance Std.Dev.  Corr
## species          (Intercept)        1.020e+00 1.0099765
##                   scale(con_dens_s)  4.693e-01 0.6850342  0.54
##                   scale(tot_dens_s)  4.486e-02 0.2117981 -0.39 -0.70
## plot:gr:loc:site (Intercept)        2.426e-01 0.4925663
## gr:loc:site      (Intercept)        1.980e-01 0.4450077
## loc:site         (Intercept)        4.294e-01 0.6552536
## site             (Intercept)        2.421e-08 0.0001556
## Number of obs: 968, groups:
## species, 26; plot:gr:loc:site, 474; gr:loc:site, 110; loc:site, 37; site, 21
##
## Conditional model:
##
##              Estimate Std. Error z value
## (Intercept)    0.574613  0.278847  2.061
## slope.degrees_s -0.190884  0.070803 -2.696
## scale(tot_dens_s) -0.140416  0.160139 -0.877
## scale(con_dens_s) -0.426845  0.201093 -2.123
## trt_II          0.549120  0.148247  3.704
## trt_FF          0.284732  0.136309  2.089
## scale(fragment.size) -0.197118  0.159596 -1.235
## scale(edge.distance) -0.068746  0.162233 -0.424
## trt_II:trt_FF -0.180307  0.213262 -0.845
## scale(tot_dens_s):trt_II 0.109923  0.193930  0.567
## scale(tot_dens_s):trt_FF 0.018673  0.152767  0.122
## scale(con_dens_s):trt_II -0.283488  0.172804 -1.641
## scale(con_dens_s):trt_FF 0.370471  0.165634  2.237
## scale(tot_dens_s):scale(fragment.size) 0.117282  0.125787  0.932
## scale(tot_dens_s):scale(edge.distance) 0.003457  0.154415  0.022
## scale(con_dens_s):scale(fragment.size) -0.208447  0.174501 -1.195
## scale(con_dens_s):scale(edge.distance) -0.063480  0.139581 -0.455
## trt_II:scale(fragment.size) -0.006105  0.138931 -0.044
## trt_II:scale(edge.distance) 0.155724  0.127403  1.222
## trt_FF:scale(fragment.size) 0.075921  0.122719  0.619
## trt_FF:scale(edge.distance) 0.130779  0.124127  1.054
```

```

## scale(fragment.size):scale(edge.distance)      0.229295    0.174775    1.312
## scale(tot_dens_s):trt_II:scale(fragment.size) -0.095562    0.256886   -0.372
## scale(tot_dens_s):trt_II:scale(edge.distance)  0.230302    0.224158    1.027
## scale(tot_dens_s):trt_FF:scale(fragment.size) -0.126785    0.235393   -0.539
## scale(tot_dens_s):trt_FF:scale(edge.distance) -0.017537    0.209178   -0.084
## scale(con_dens_s):trt_II:scale(fragment.size) -0.121967    0.232343   -0.525
## scale(con_dens_s):trt_II:scale(edge.distance) -0.096600    0.166590   -0.580
## scale(con_dens_s):trt_FF:scale(fragment.size)  0.646466    0.215039    3.006
## scale(con_dens_s):trt_FF:scale(edge.distance) -0.194816    0.168200   -1.158
##
## Pr(>|z|)
## (Intercept)                                0.039334 *
## slope.degrees_s                            0.007018 **
## scale(tot_dens_s)                           0.380575
## scale(con_dens_s)                           0.033785 *
## trt_II                                       0.000212 ***
## trt_FF                                       0.036719 *
## scale(fragment.size)                        0.216791
## scale(edge.distance)                        0.671750
## trt_II:trt_FF                               0.397848
## scale(tot_dens_s):trt_II                    0.570838
## scale(tot_dens_s):trt_FF                    0.902718
## scale(con_dens_s):trt_II                    0.100897
## scale(con_dens_s):trt_FF                    0.025307 *
## scale(tot_dens_s):scale(fragment.size)      0.351139
## scale(tot_dens_s):scale(edge.distance)       0.982137
## scale(con_dens_s):scale(fragment.size)       0.232271
## scale(con_dens_s):scale(edge.distance)       0.649261
## trt_II:scale(fragment.size)                 0.964948
## trt_II:scale(edge.distance)                 0.221595
## trt_FF:scale(fragment.size)                 0.536146
## trt_FF:scale(edge.distance)                 0.292068
## scale(fragment.size):scale(edge.distance)    0.189539
## scale(tot_dens_s):trt_II:scale(fragment.size) 0.709893
## scale(tot_dens_s):trt_II:scale(edge.distance) 0.304228
## scale(tot_dens_s):trt_FF:scale(fragment.size) 0.590156
## scale(tot_dens_s):trt_FF:scale(edge.distance) 0.933187
## scale(con_dens_s):trt_II:scale(fragment.size) 0.599623
## scale(con_dens_s):trt_II:scale(edge.distance) 0.562006
## scale(con_dens_s):trt_FF:scale(fragment.size) 0.002645 **
## scale(con_dens_s):trt_FF:scale(edge.distance) 0.246767
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

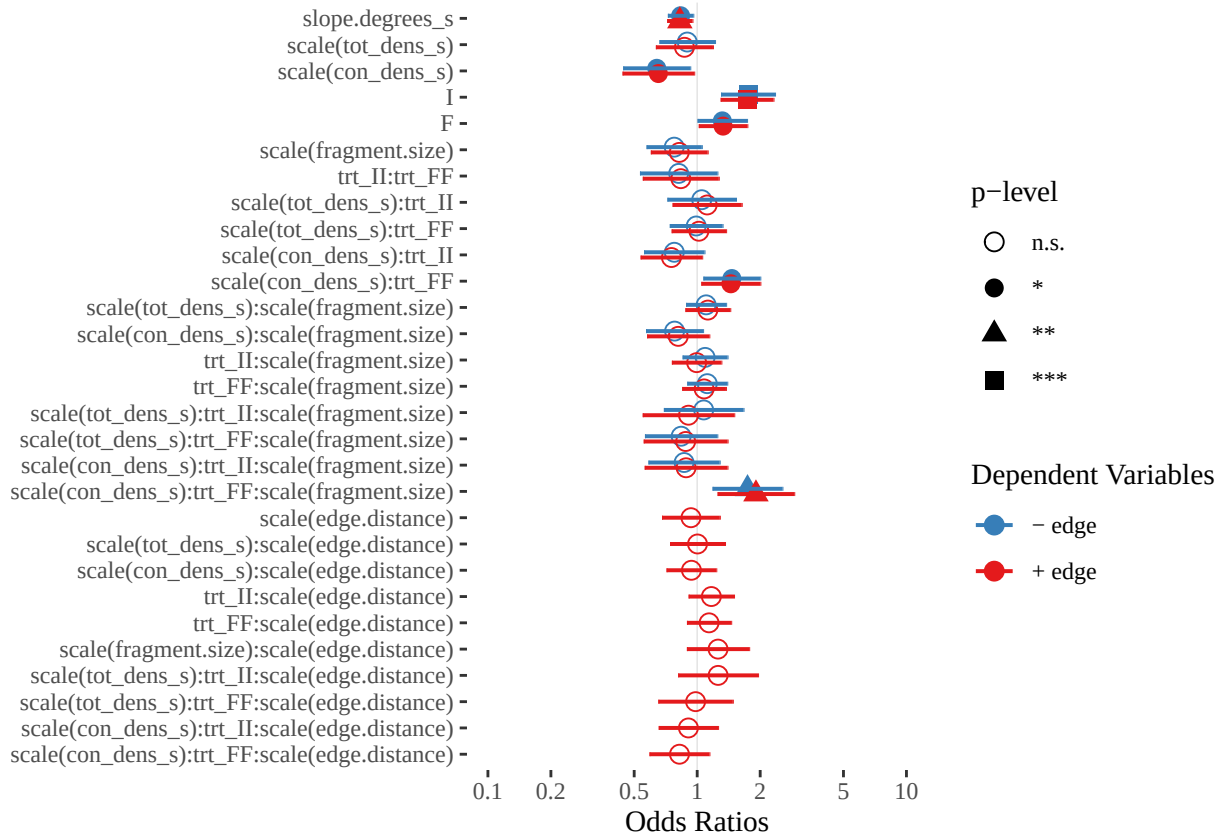
## what about effects on fragment area effects. Comparing tree plots of
## model with and without edge effects

```

```

plot_models(m_cdd_s_ris, m_cdde_s_ris,
            m.labels = c("- edge", "+ edge"), p.shape=TRUE)

```



- None of the edge effects are statistically significant (the code pushes them to the bottom)
- the effects of including edge distance on other parameter estimates are slight. Notice that statistical significance never switches, but more importantly, the effects on magnitude are trivial.

### 3.1.5 Models split by categorical fragment area

The effect of fragment area is complicated, so looking at separate models for different size-classes of fragments.

```
# Using 3 categories
sdls <- mutate(sdls,
  frag_sizeclass3 = cut(fragment.size,
    round(quantile(site_dat$fragment.size,
      c(0, 0.33, 0.66, 1))),
    include.lowest=TRUE))

## quick look at how many fragments and seedlings within each
## fragment size category
sdls |> group_by(frag_sizeclass3) |> summarise(n_frag = n_distinct(site),
  n_sdl = n())
```

```
## # A tibble: 3 x 3
##   frag_sizeclass3 n_frag n_sdl
##   <fct>          <int> <int>
## 1 [1,10]         11    260
```

```
## 2 (10,59]          6   331
## 3 (59,149]        4   377
```

```
m_cdd_s_ris_frag3 <- lapply(split(sdls, f= sdls$frag_sizeclass3), function(d){
  var_d <- d |> group_by(species) |>
    summarise(var_dens = var(con_dens_s)) |>
    filter(!is.na(var_dens) & var_dens > 0)

  glmmTMB(Pr_s ~ slope.degrees_s +
    trt_I:trt_F +
    (scale(tot_dens_s) + scale(con_dens_s)) *
    (trt_I + trt_F) +
    (scale(con_dens_s) + scale(tot_dens_s)||species) +
    (1|site/loc/gr/plot),
    weights = census.start,
    data = filter(d, species %in% var_d$species),
    family=binomial)})

term_nms <- names(fixef(m_cdd_s_ris_frag3[[1]])$cond)
names(m_cdd_s_ris_frag3) <- c("1 - 4", "5 - 32", "33 - 150") ## renaming to
## be clearer and correct (from looking at observed boundaries)

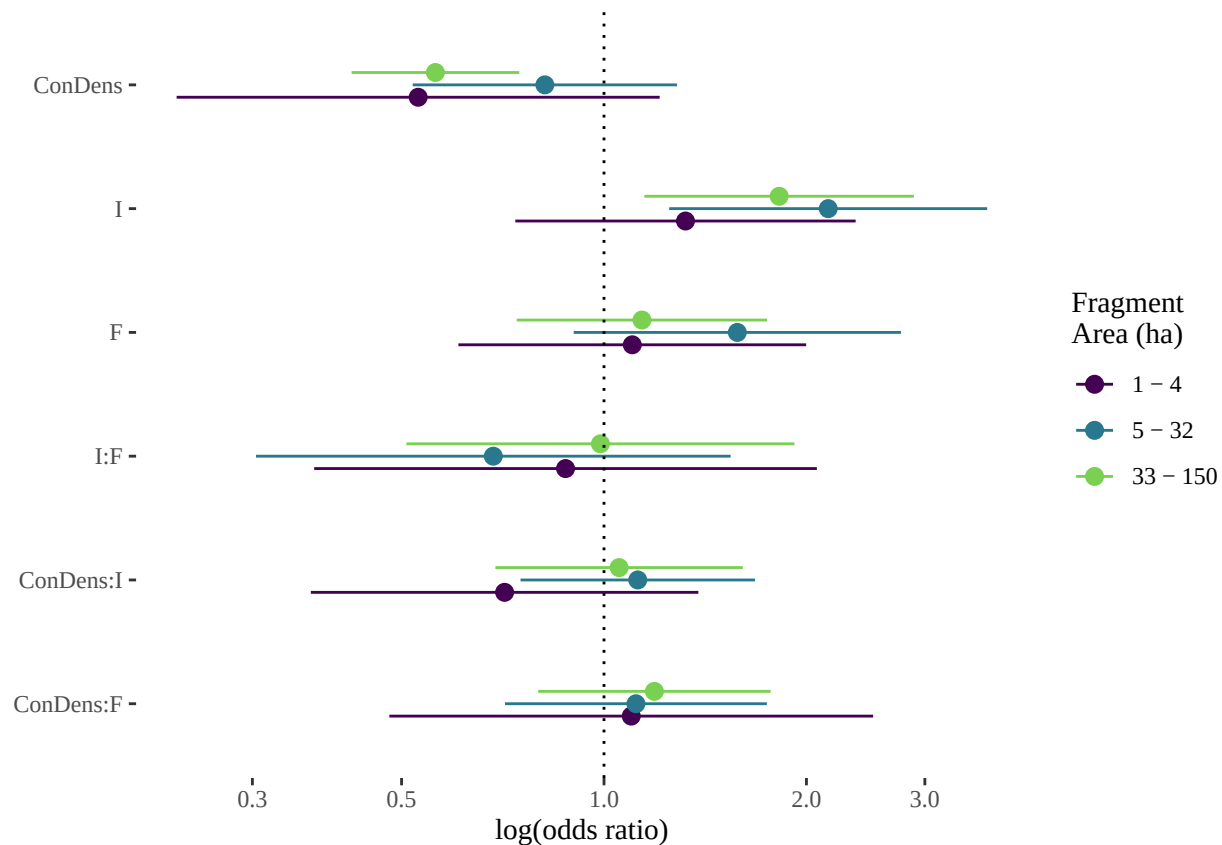
plot_discrete <- plot_models(m_cdd_s_ris_frag3,
  rm.terms=c("slope.degrees_s",
    term_nms[str_detect(term_nms, "tot")] ), ## declutter
  m.labels=names(m_cdd_s_ris_frag3), p.shape=TRUE,
  show.values = TRUE) +
  labs(colour = "Fragment Area")

pl_discrete <- plot_discrete$data |>
  mutate(term = str_replace_all(term, fixed("scale(con_dens_s)"), "ConDens"),
    term = str_replace_all(term, "trt_[FI]", ""),
    term = factor(term, levels = rev(unique(term))),
    FragArea = factor(group, levels = c("1 - 4", "5 - 32", "33 - 150")))

pl_discrete <- ggplot(pl_discrete, aes(x = estimate, xmin = conf.low, xmax = conf.high,
  y = term, colour = FragArea)) +
  geom_pointrange(position = position_dodge2(width = 0.3)) +
  geom_vline(xintercept=1, linetype = "dotted") +
  scale_x_continuous(trans = "log10", n.breaks = 7) +
  scale_colour_viridis_d(option="D", end=0.8) +
  labs(x = "log(odds ratio)", y = NULL, colour = "Fragment\nArea (ha)")

pl_discrete
```





```
ggsave(pl_discrete, file = "figures/CatFragPlot.png", height = 5, width = 5)
ggsave(pl_discrete, file = "figures/CatFragPlot.pdf", height = 5, width = 5)
```

```
map(m_cdd_s_ris_frag3, summary)
```

```
## $'1 - 4'
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) *
## (trt_I + trt_F) + (scale(con_dens_s) + scale(tot_dens_s) ||
## species) + (1 | site/loc/gr/plot)
## Data: filter(d, species %in% var_d$species)
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
##    647.9    711.0   -305.9   611.9     228
##
## Random effects:
##
## Conditional model:
## Groups      Name              Variance Std.Dev. Corr
## species     (Intercept)        1.83686  1.3553
##              scale(con_dens_s) 0.26340  0.5132  0.00
##              scale(tot_dens_s) 0.03298  0.1816  0.00 0.00
## plot:gr:loc:site (Intercept)    0.37706  0.6141
```

```

## gr:loc:site      (Intercept)      0.22788  0.4774
## loc:site         (Intercept)      0.40536  0.6367
## site             (Intercept)      0.40536  0.6367
## Number of obs: 246, groups:
## species, 15; plot:gr:loc:site, 138; gr:loc:site, 32; loc:site, 11; site, 11
##
## Conditional model:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.287783    0.549560   2.343  0.01911 *
## slope.degrees_s  -0.353985    0.130779  -2.707  0.00679 **
## scale(tot_dens_s)  0.003197    0.227034   0.014  0.98877
## scale(con_dens_s) -0.636388    0.421941  -1.508  0.13149
## trt_II            0.279039    0.297271   0.939  0.34790
## trt_FF            0.096783    0.303751   0.319  0.75001
## trt_II:trt_FF     -0.131722    0.439076  -0.300  0.76418
## scale(tot_dens_s):trt_II  0.425125    0.294966   1.441  0.14951
## scale(tot_dens_s):trt_FF -0.143306    0.461595  -0.310  0.75621
## scale(con_dens_s):trt_II -0.340369    0.338505  -1.006  0.31465
## scale(con_dens_s):trt_FF  0.093301    0.422564   0.221  0.82525
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $'5 - 32'
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) *
##       (trt_I + trt_F) + (scale(con_dens_s) + scale(tot_dens_s) ||
##       species) + (1 | site/loc/gr/plot)
## Data: filter(d, species %in% var_d$species)
## Weights: census.start
##
##      AIC      BIC    logLik deviance df.resid
##    664.1    731.5   -314.0    628.1     294
##
## Random effects:
##
## Conditional model:
## Groups      Name              Variance Std.Dev.  Corr
## species      (Intercept)      1.318e+00 1.148e+00
##               scale(con_dens_s) 1.120e-01 3.347e-01 0.00
##               scale(tot_dens_s) 4.236e-13 6.508e-07 0.00 0.00
## plot:gr:loc:site (Intercept)  3.359e-01 5.796e-01
## gr:loc:site      (Intercept)  7.463e-02 2.732e-01
## loc:site         (Intercept)  1.080e-01 3.287e-01
## site            (Intercept)  2.001e-08 1.414e-04
## Number of obs: 312, groups:
## species, 18; plot:gr:loc:site, 160; gr:loc:site, 39; loc:site, 13; site, 6
##
## Conditional model:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.24674    0.37165   0.664  0.50674
## slope.degrees_s  -0.02479    0.14485  -0.171  0.86409
## scale(tot_dens_s)  0.03495    0.24819   0.141  0.88802
## scale(con_dens_s) -0.20248    0.23091  -0.877  0.38055

```

```

## trt_II                0.76762      0.27776    2.764  0.00572 **
## trt_FF                0.45647      0.28584    1.597  0.11028
## trt_II:trt_FF        -0.37904      0.41453   -0.914  0.36051
## scale(tot_dens_s):trt_II -0.48810    0.23209   -2.103  0.03546 *
## scale(tot_dens_s):trt_FF  0.07187    0.24492    0.293  0.76919
## scale(con_dens_s):trt_II  0.11561    0.20483    0.564  0.57248
## scale(con_dens_s):trt_FF  0.10930    0.22873    0.478  0.63275
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $'33 - 150'
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) *
##      (trt_I + trt_F) + (scale(con_dens_s) + scale(tot_dens_s) ||
##      species) + (1 | site/loc/gr/plot)
## Data: filter(d, species %in% var_d$species)
## Weights: census.start
##
##      AIC      BIC    logLik deviance df.resid
##    876.7    946.6   -420.3    840.7     342
##
## Random effects:
##
## Conditional model:
## Groups      Name              Variance Std.Dev.  Corr
## species      (Intercept)       9.996e-01 9.998e-01
##               scale(con_dens_s) 3.124e-11 5.589e-06 0.00
##               scale(tot_dens_s) 2.729e-10 1.652e-05 0.00 0.00
## plot:gr:loc:site (Intercept)    2.885e-01 5.371e-01
## gr:loc:site      (Intercept)    4.468e-01 6.684e-01
## loc:site         (Intercept)    2.044e-01 4.521e-01
## site            (Intercept)    3.670e-09 6.058e-05
## Number of obs: 360, groups:
## species, 16; plot:gr:loc:site, 166; gr:loc:site, 39; loc:site, 13; site, 4
##
## Conditional model:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.75266    0.36370   2.069   0.0385 *
## slope.degrees_s -0.11265    0.13185  -0.854   0.3929
## scale(tot_dens_s) -0.13672    0.18415  -0.742   0.4578
## scale(con_dens_s) -0.57719    0.14643  -3.942 8.09e-05 ***
## trt_II          0.59971    0.23542   2.547   0.0109 *
## trt_FF          0.12997    0.21867   0.594   0.5523
## trt_II:trt_FF   -0.01226    0.33895  -0.036   0.9711
## scale(tot_dens_s):trt_II 0.11552    0.32481   0.356   0.7221
## scale(tot_dens_s):trt_FF 0.01290    0.24027   0.054   0.9572
## scale(con_dens_s):trt_II 0.05161    0.21609   0.239   0.8112
## scale(con_dens_s):trt_FF 0.17254    0.20290   0.850   0.3951
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

### 3.1.6 Take-homes

The model gets a bit complex when split like this, but it serves as a useful exploration of what's going on.

- Conspecific density reduces survival most clearly in the larger fragments.
- The results of fungicide:cdens make a bit more sense when the data are broken up like this. Fungicide only interacts with cdens in the largest fragments, which makes sense, because these are the fragments where cdens has an effect.
- insects seem to only reduce plant survival in the medium and large fragments

### 3.1.7 Graphics

```
## refitting model to control order of coefficients
m_cdd_s_ris <- glmmTMB(Pr_s ~ slope.degrees_s +
  (scale(tot_dens_s) + scale(con_dens_s)) +
  scale(fragment.size) +
  trt_I*trt_F +
  (scale(tot_dens_s) + scale(con_dens_s)) *
  (trt_I + trt_F) *
  scale(fragment.size) +
  (scale(con_dens_s) + scale(tot_dens_s)|species) +
  ## setting cor to 0 to converge
  (1|site/loc/gr/plot),
  weights = census.start,
  data = sdls,
  family=binomial)

labs <- c("ConDens", "FragArea", "I", "F", "I:F",
  "ConDens:I", "ConDens:F", "FragArea:\n ConDens",
  "FragArea:I", "FragArea:F",
  "FragArea:\n (ConDens:I)", "FragArea:\n (ConDens:F)")

tp_cdd_s_ris <-
  tidy(m_cdd_s_ris, conf.int = TRUE) |>
  filter(effect == "fixed",
    !str_detect(term, "tot_dens"),
    !term %in% c("(Intercept)", "slope.degrees_s")) |>
  mutate(labels = factor(labs, levels = rev(labs)),
    Biocide = case_when(
      str_detect(term, "trt_II") ~ "Insecticide",
      str_detect(term, "trt_FF") ~ "Fungicide",
      .default = "Control"),
    Biocide = factor(ifelse(str_detect(term, "trt_II:trt_FF"),
      "Both", Biocide),
      levels = c("Control", "Insecticide",
        "Fungicide", "Both"))) |>
  ggplot(aes(y = labels, x = estimate, xmin = conf.low, xmax = conf.high,
    colour = Biocide)) +
  geom_pointrange(aes(shape = p.value < 0.05 )) +
  geom_vline(xintercept=0, linetype = "dotted") +
  scale_colour_brewer(palette="Set2") +
```

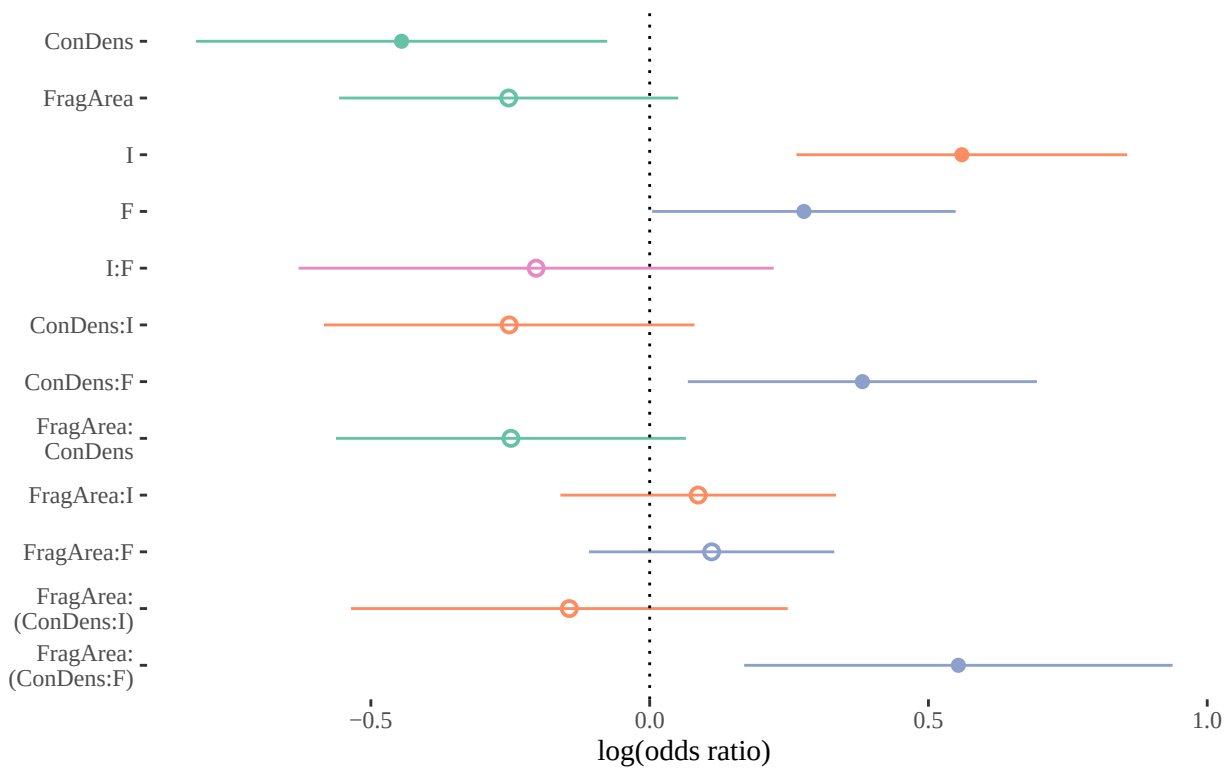
```

scale_shape_manual(values=c(1, 16), guide = "none" ) +
labs(y = NULL, x = "log(odds ratio)" ) +
guides(colour = guide_legend(title= "Biocide:", position = "top",
                             direction = "horizontal",
                             title.position = "left", title.hjust = 0.5)) +

theme(
  legend.margin = margin(0, 0, 0, 0),
  legend.justification.top = "left",
  legend.location = "plot",
  plot.title.position = "plot"
)
tp_cdd_s_ris

```

Biocide: ● Control ● Insecticide ● Fungicide ● Both



```

p_cdd_s_ris <- predict_response(m_cdd_s_ris,
                              terms=c("con_dens_s[1:80, by = 1]",
                                       "trt_I", "trt_F",
                                       "fragment.size[5, 65, 125]"))

pr_cdd_s_ris <- residualize_over_grid(p_cdd_s_ris, m_cdd_s_ri) |>
  bind_cols(species = sdls$species) |>
  mutate(trt = factor(
    case_when(
      group == "0" & facet == "0" ~ "Control",
      group == "0" & facet == "F" ~ "Fungicide",
      group == "I" & facet == "0" ~ "Insecticide",

```

```

    group == "I" & facet == "F" ~ "Both"),
    levels = c("Control", "Insecticide", "Fungicide", "Both")),
    facet_lab = "Fragment Area (ha)" )>
rename("con_dens" = "x", "frag_area" = "panel" )>
group_by(con_dens, trt, frag_area, facet_lab, species) |>
summarise(predicted = mean(predicted), n = n())

```

## 'summarise()' has grouped output by 'con\_dens', 'trt', 'frag\_area',  
 ## 'facet\_lab'. You can override using the '.groups' argument.

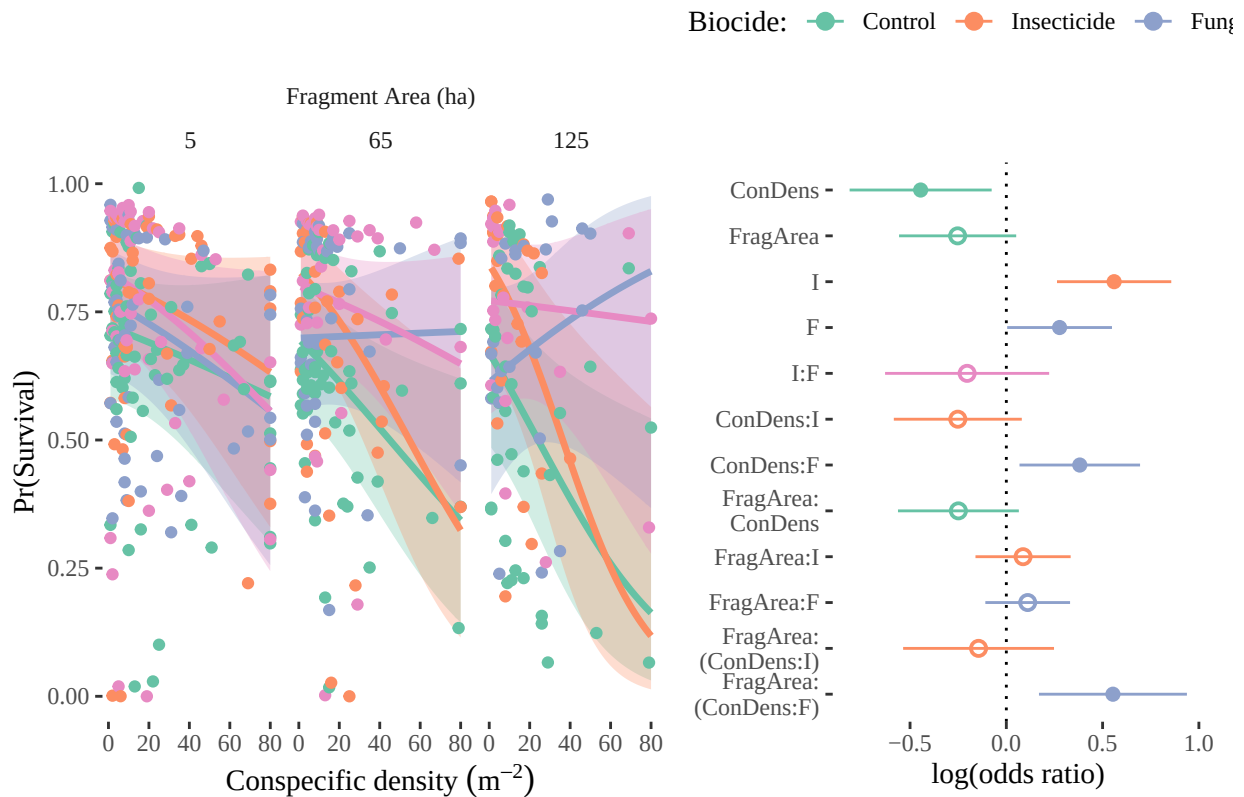
```

p_cdd_s_ris <- as.data.frame(p_cdd_s_ris) |>
mutate(trt = factor(
  case_when(
    group == "0" & facet == "0" ~ "Control",
    group == "0" & facet == "F" ~ "Fungicide",
    group == "I" & facet == "0" ~ "Insecticide",
    group == "I" & facet == "F" ~ "Both"),
  levels = c("Control", "Insecticide", "Fungicide", "Both")),
  facet_lab = "Fragment Area (ha)" )>
rename("con_dens" = "x", "frag_area" = "panel")

pl_cdd_s <- ggplot(p_cdd_s_ris,
  aes(x=con_dens, y = predicted, colour = trt)) +
  ggh4x::facet_nested(~facet_lab + frag_area) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high, fill = trt),
    colour = NA, alpha = 0.3) +
  geom_line(linewidth = 1.2) +
  geom_point(data = pr_cdd_s_ris) +
  scale_colour_brewer(palette="Set2") +
  scale_fill_brewer(palette="Set2", guide = "none") +
  labs(x = expression(Conspecific~density~(m^-2)), y = "Pr(Survival)",
    colour = "Treatment") + theme(legend.position="none")

mod_plot <- (pl_cdd_s | tp_cdd_s_ris ) +
  plot_layout(widths=c(0.6, 0.4))
mod_plot

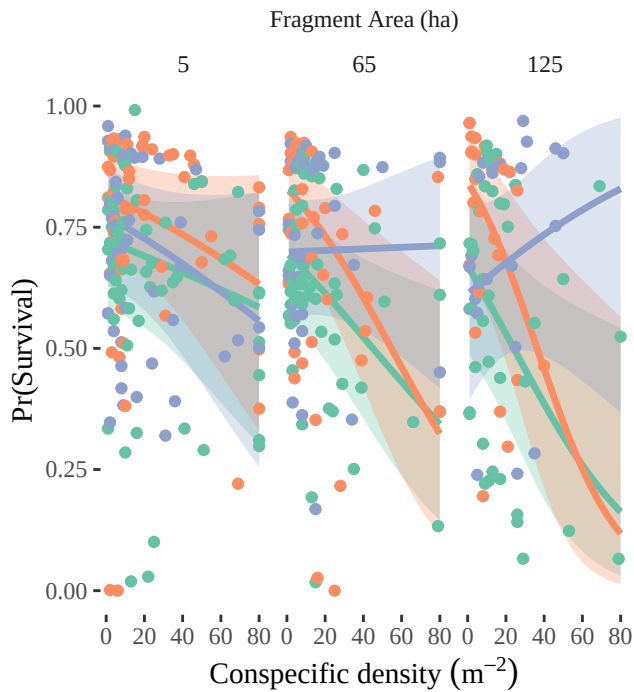
```



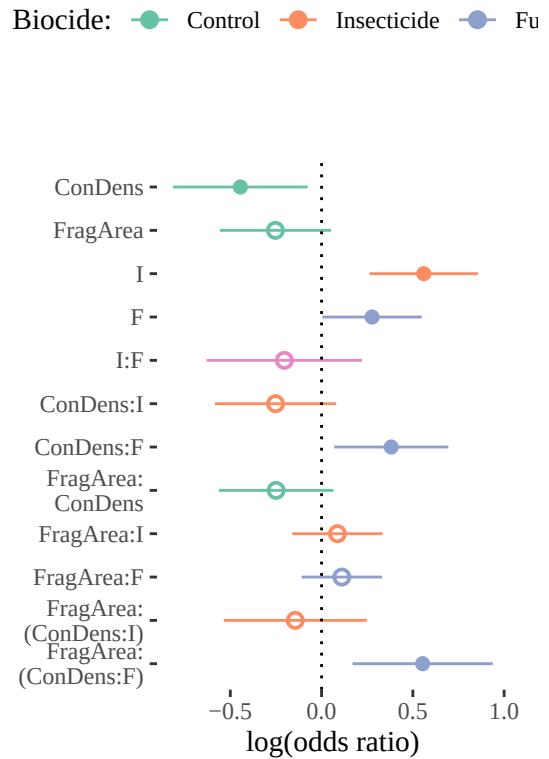
```
# the predictions with the "both" treatment get really busy. Given the
# lack of an interaction, trying without "both"
pl_cdd_s_2 <- ggplot(filter(p_cdd_s_ris, trt != "Both"),
  aes(x=con_dens, y = predicted, colour = trt)) +
  ggh4x::facet_nested(~facet_lab + frag_area) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high, fill = trt),
    colour = NA, alpha = 0.3) +
  geom_line(linewidth = 1.2) +
  geom_point(data = filter(pr_cdd_s_ris, trt != "Both")) +
  scale_colour_brewer(palette="Set2") +
  scale_fill_brewer(palette="Set2", guide = "none") +
  labs(x = expression(Conspecific-density~(m^-2)), y = "Pr(Survival)",
    colour = "Treatment") + theme(legend.position="none")

mod_plot2 <- (pl_cdd_s_2 | tp_cdd_s_ris ) +
  plot_layout(widths=c(0.6, 0.4)) + plot_annotation(tag_level = "A") ## better
mod_plot2
```

A



B



```
ggsave(mod_plot2, file = "figures/model_results.png", height = 5.5, width = 7.5)
ggsave(mod_plot2, file = "figures/model_results.pdf", height = 5.5, width = 7.5)
```

### 3.1.8 Plotting effects of fragment area

The effects are complex and hinge on a non-significant interaction between fragment area and the fungicide effect. To interrogate them more, here are some plots (basically Neyman-Johnson interaction plots)

```
## plot the effect of density over fragment size
## helps to scale and log data outside the model formula to set this up
int_data <- sdls |>
  mutate(con_dens_ss = scale(con_dens_s),
         tot_dens_ss = scale(tot_dens_s),
         fragsize_s = scale(fragment.size))

int_mod <- glmmTMB(Pr_s ~ slope.degrees_s +
  trt_I:trt_F +
  (tot_dens_ss + con_dens_ss) *
  (trt_I + trt_F) * fragsize_s +
  (1 + tot_dens_ss + con_dens_ss | species) +
  (1 | site/loc/gr/plot),
  weights = census.start, family = binomial,
  data = int_data)

preddat <- expand.grid(
```



```

trt_F = levels(int_data$trt_F),
trt_I = levels(int_data$trt_I),
con_dens_ss = 1,
fragsize_s = seq(-1, 2.1, length = 20)) |>
filter(!(trt_F == "F" & trt_I == "I"))

xmat <- model.matrix(~ con_dens_ss + con_dens_ss:(trt_F + trt_I + fragsize_s) +
                     con_dens_ss:trt_F:fragsize_s +
                     con_dens_ss:trt_I:fragsize_s,
                     preddat)[ , -1]

preddat$int_hat <- as.vector(xmat %*% fixef(int_mod)$cond[colnames(xmat)])
vmat <- (vcov(int_mod)$cond[colnames(xmat), colnames(xmat)])
preddat$int_se <- sqrt(diag(xmat %*% vmat %*% t(xmat)))

preddat <- preddat |> mutate(.lower = int_hat - 1.96*int_se,
                           .upper = int_hat + 1.96*int_se,
                           frag_area = fragsize_s*
                             sd(sdls$fragment.size) +
                             mean(sdls$fragment.size),
                           trt = factor(case_when(
                             trt_F == "O" & trt_I == "O" ~ "Control",
                             trt_F == "F" & trt_I == "O" ~ "Fungicide",
                             trt_F == "O" & trt_I == "I" ~ "Insecticide"))))

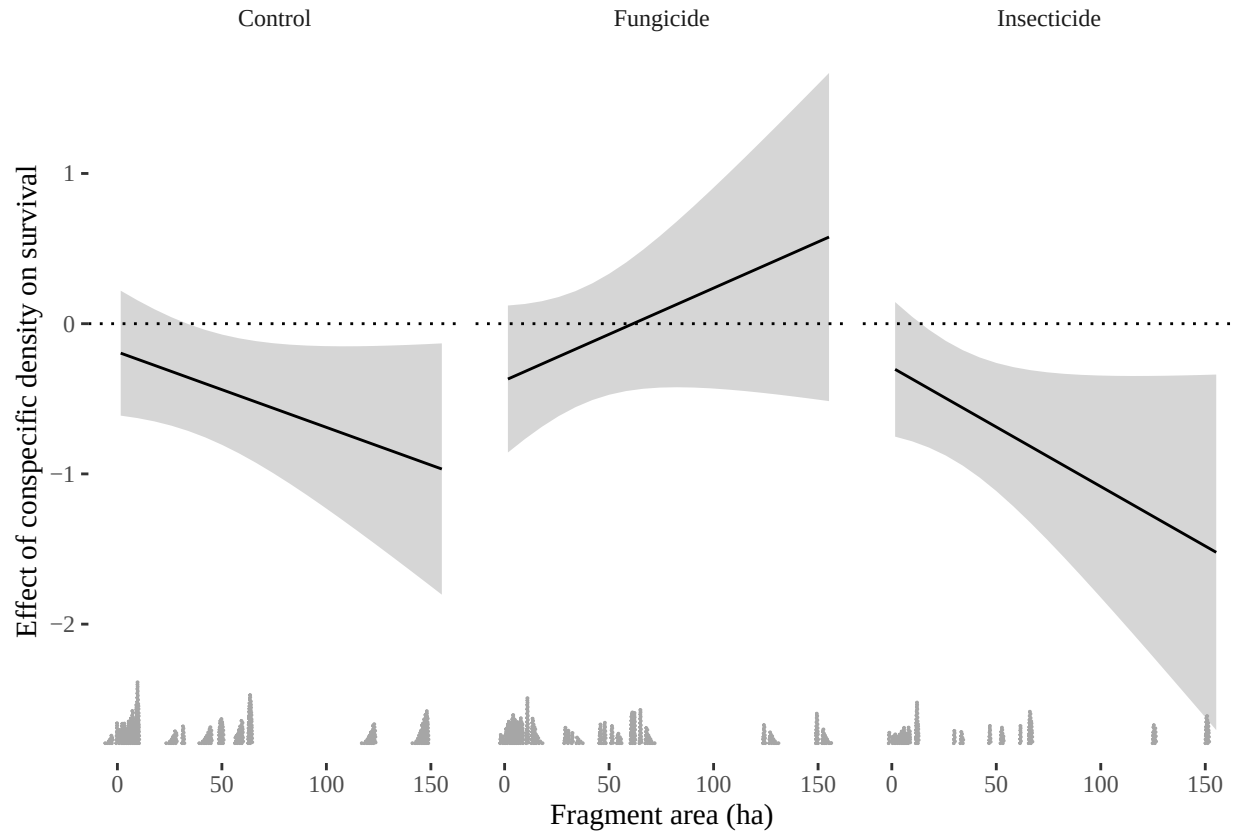
int_plot <- ggplot(preddat, aes(x = frag_area)) + facet_wrap( ~ trt) +
  geom_ribbon(aes(y = int_hat, ymin = .lower, ymax = .upper),
            alpha = 0.2, colour = NA) +
  geom_line(aes(y = int_hat)) + geom_hline(yintercept=0, linetype = "dotted") +
  labs(x = "Fragment area (ha)",
       y = "Effect of conspecific density on survival")

sdls <- mutate(sdls, trt = case_when(
  trt_F == "F" ~ "F",
  trt_I == "I" ~ "I",
  .default = "C"),
  trt = factor(trt, labels = c("Control", "Fungicide", "Insecticide")))

int_plot <- int_plot +
  ggdist::geom_dots(data = sdls, aes(x = fragment.size), y = -2.8,
                    smooth = ggdist::smooth_unbounded(), layout = "swarm",
                    side = "top", binwidth = 1, # alpha = 0.7,
                    overflow = "compress")

int_plot

```



```
ggsave(int_plot, file = "figures/interaction_plot.png", width=6.6, height = 4)
ggsave(int_plot, file = "figures/interaction_plot.pdf", width=6.6, height = 4)
```

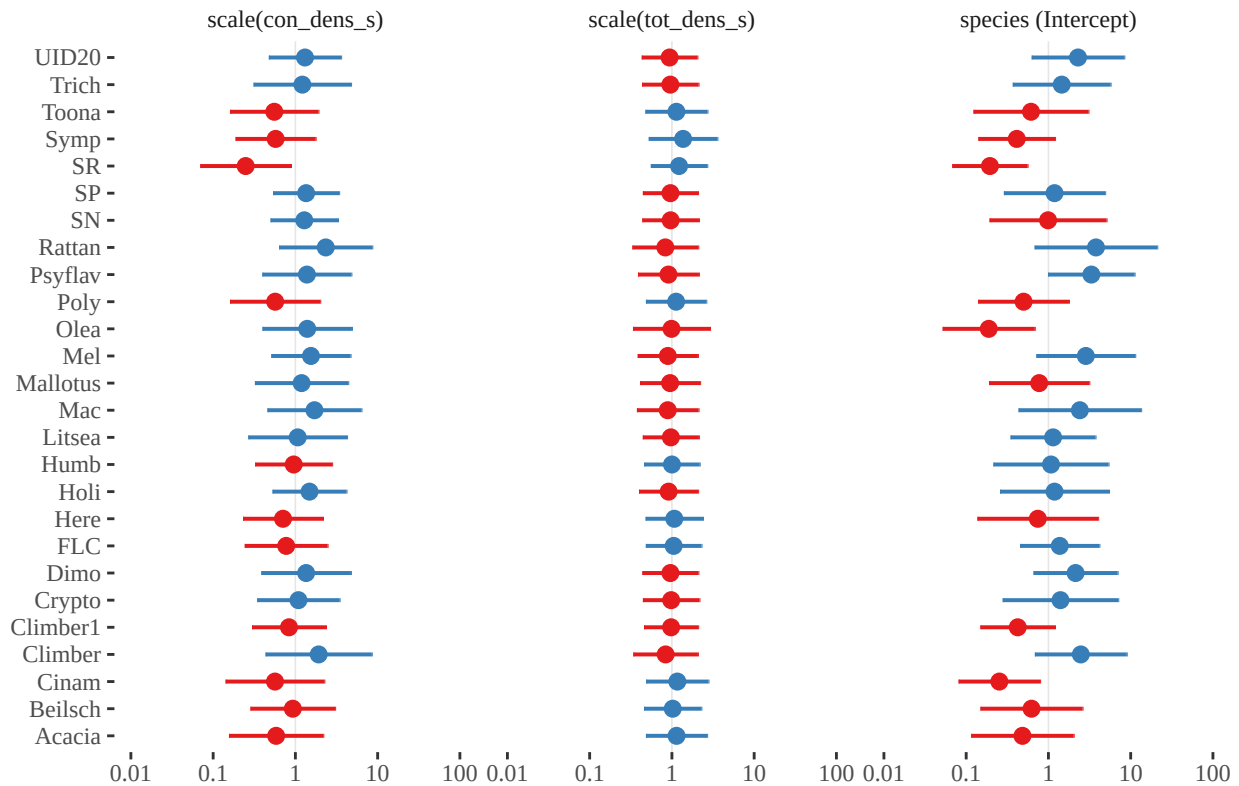
### 3.2 Species specific inferences.

Initial analyses suggested that some species were heavily influencing the results.

The random slope should partly account for that.

```
sjPlot::plot_model(m_cdd_s_ris, type = "re", terms = "species", ri.nr = 1)
```

## Random effects



```
## observable variation among species in CDD slopes, but not in tot_dens_s
sp_eff <- as.data.frame(ranef(m_cdd_s_ris, condVar = T)$cond$species)
```

```
sp_eff <- bind_cols(sp_eff,
  t(apply(attr(sp_eff, "condVar"), 3, function(x)
    sqrt(diag(x))))))
```

```
## New names:
## * ' ' -> '...4'
## * ' ' -> '...5'
## * ' ' -> '...6'
```

```
names(sp_eff) <- c("Intercept", "con_dens", "tot_dens",
  "Intercept_se", "con_dens_se", "tot_dens_se")
```

```
## extract total effects
```

```
sp_eff <- mutate(sp_eff, sp_names = row.names(sp_eff),
  Intercept = Intercept + fixef(m_cdd_s_ris)$cond[1],
  con_dens = con_dens +
    fixef(m_cdd_s_ris)$cond["scale(con_dens_s)"],
  tot_dens = tot_dens +
    fixef(m_cdd_s_ris)$cond["scale(tot_dens)"])
```

```
sp_eff <- left_join(sp_eff, sp_codes, by = c("sp_names" = "code"))
## wrangle species names
```

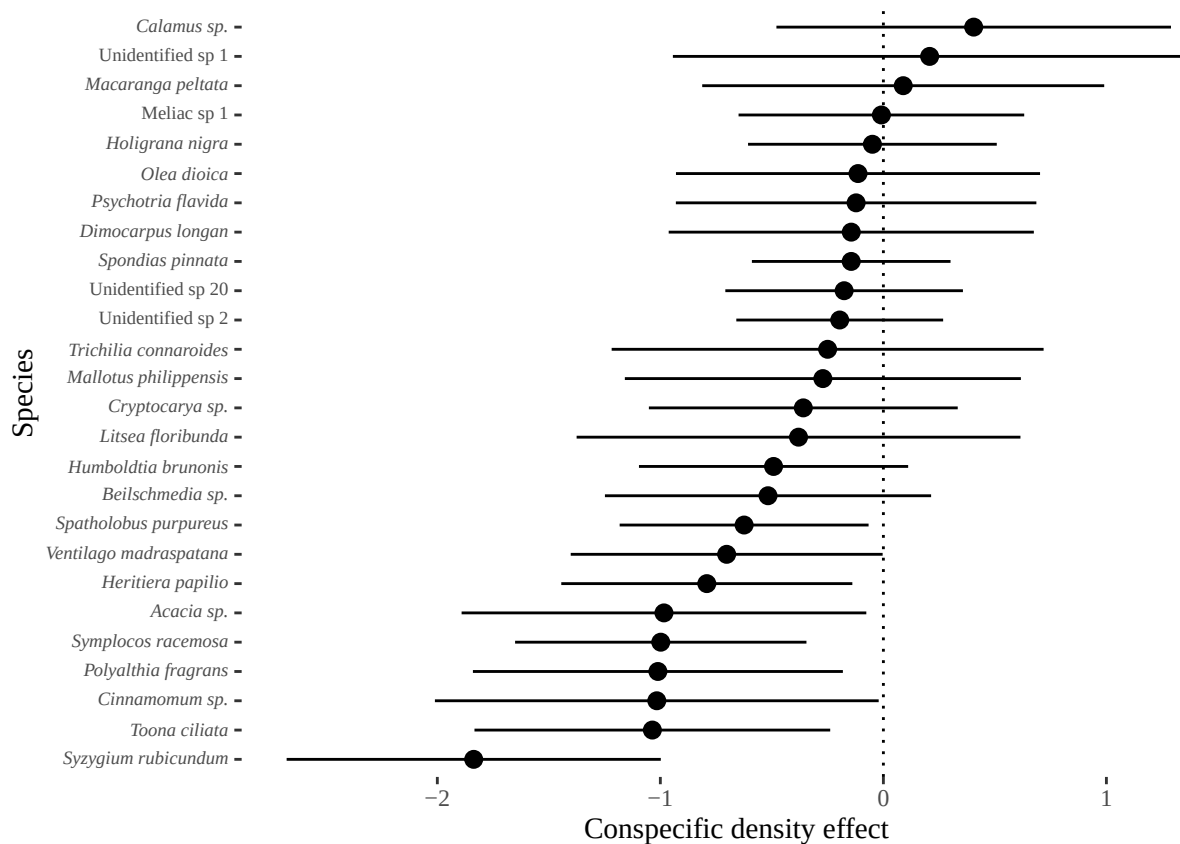
```

sp_eff <- sp_eff |> mutate(spbins = case_when(
  genus %in% c("Unidentified", "Meliac") ~ paste(genus, species, sep = "~"),
  genus != "Unidentified" & str_starts("sp", species) ~
    paste(paste0("italic('", genus, "'"), species, sep = "~"),
    paste0("italic('", paste(genus, species, sep = " "), "'"))))

blup_plot <- sp_eff |> arrange(con_dens) |>
  ggplot(aes(y = reorder(spbins, con_dens), x = con_dens,
    xmin = con_dens - 2*con_dens_se,
    xmax = con_dens + 2*con_dens_se)) +
  geom_vline(xintercept=0, linetype = "dotted") +
  geom_pointrange() +
  scale_y_discrete(labels = scales::label_parse()) +
  labs(x = "Conspecific density effect", y = "Species")

blup_plot <- blup_plot + theme(axis.text.y = element_text(size = 7))
blup_plot

```



```
# ggsave(blup_plot, file = "figures/cdd_blups.png", width = 4, height = 5)
```

```
anova(m_cdd_s_ris, m_cdd_s_ri)
```

```
## Data: sdfs
```

```
## Models:
```

```
## m_cdd_s_ri: Pr_s ~ slope.degrees_s + trt_I:trt_F + (scale(tot_dens_s) + scale(con_dens_s)) * , zi=~0
```

```
## m_cdd_s_ri:      (trt_I + trt_F) * scale(fragment.size) + (1 | species) + , zi=~0, disp=~1
## m_cdd_s_ri:      (1 | site/loc/gr/plot), zi=~0, disp=~1
## m_cdd_s_ris: Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) + , zi=~0, disp=~1
## m_cdd_s_ris:      scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) + , zi=~0, disp=~1
## m_cdd_s_ris:      scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) + , zi=~0, disp=~1
## m_cdd_s_ris:      (scale(con_dens_s) + scale(tot_dens_s) | species) + (1 | , zi=~0, disp=~1
## m_cdd_s_ris:      site/loc/gr/plot), zi=~0, disp=~1
##           Df      AIC      BIC logLik deviance Chisq Chi Df Pr(>Chisq)
## m_cdd_s_ri 25 2232.1 2353.9 -1091.0 2182.1
## m_cdd_s_ris 30 2215.1 2361.3 -1077.5 2155.1 27.02      5 5.654e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Substantial variation in cdd among species -

Perhaps remove most abundant species to check if patterns are robust.

```
sdl$ |> group_by(species) |> summarise(n = sum(census.start)) |>
  arrange(desc(n)) |> print(n = 26)
```

```
## # A tibble: 26 x 2
##   species      n
##   <fct>    <int>
## 1 SR        3598
## 2 Symp       636
## 3 Climber1   627
## 4 FLC        313
## 5 Cinam      137
## 6 Psyflav    116
## 7 Dimo        99
## 8 SP          85
## 9 Litsea      72
## 10 Olea       71
## 11 UID20       65
## 12 Climber     59
## 13 Poly        54
## 14 Mel         43
## 15 Beilsch     42
## 16 Holi        34
## 17 Toona       28
## 18 SN          25
## 19 Trich       22
## 20 Mallotus    19
## 21 Acacia      16
## 22 Mac         12
## 23 Rattan      11
## 24 Humb        9
## 25 Crypto       8
## 26 Here        7
```

```
sp_common <- sdl$ |> group_by(species) |> summarise(n = sum(census.start)) |>
  arrange(desc(n)) |> pull(species)
sp_mods <- lapply(c("none", as.character(sp_common[1:5])), function(i) {
  update(m_cdd_s_ris, data = filter(sdl$, !species == i))})
```

```

## Warning in (function (start, objective, gradient = NULL, hessian = NULL, :
## NA/NaN function evaluation
## Warning in (function (start, objective, gradient = NULL, hessian = NULL, :
## NA/NaN function evaluation
## Warning in (function (start, objective, gradient = NULL, hessian = NULL, :
## NA/NaN function evaluation

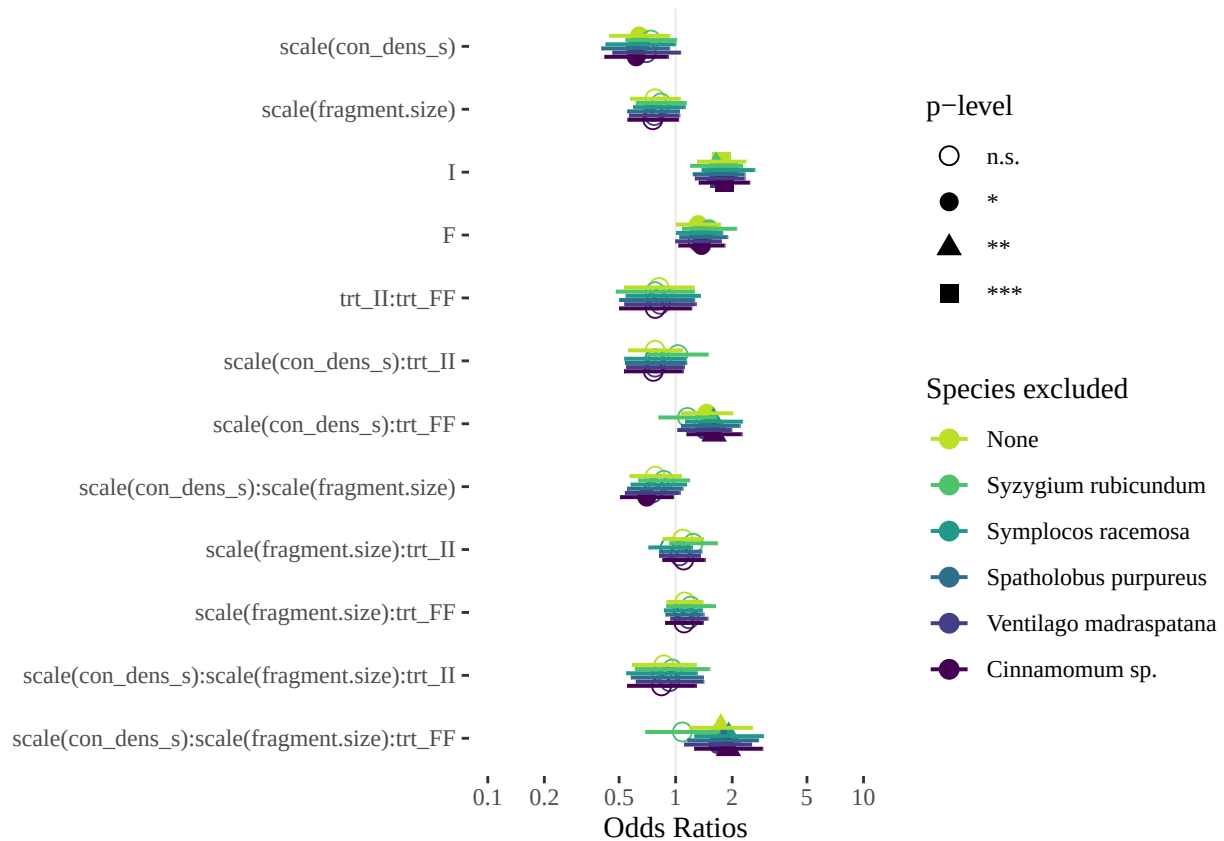
## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; non-positive-definite Hessian matrix. See vignette('troubleshooting')

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; singular convergence (7). See vignette('troubleshooting'),
## help('diagnose')

sp_codes <- mutate(sp_codes, spbin = paste(genus, species))
term_nms <- names(fixef(sp_mods[[1]])$cond)
names(sp_mods) <- c("None", as.character(sp_common[1:5]))
plot_models(sp_mods,
  rm.terms = c("slope.degrees_s",
    term_nms[str_detect(term_nms, "tot")] ),
  m.labels = c("None",
    sp_codes$spbin[match(sp_common[1:5], sp_codes$code)] ),
  p.shape=TRUE) + labs(colour = "Species excluded") +
  scale_color_viridis_d(end=0.9) ## getting rid of the awful yellow.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```



```
map(sp_mods, summary)
```

```
## $None
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##       scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##       (scale(con_dens_s) + scale(tot_dens_s) | species) + (1 |
##       site/loc/gr/plot)
## Data: filter(sdls, !species == i)
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
##  2215.0   2361.3 -1077.5   2155.0     938
##
## Random effects:
##
## Conditional model:
## Groups      Name              Variance Std.Dev.  Corr
## species      (Intercept)        1.027e+00 1.0131846
##               scale(con_dens_s)  3.887e-01 0.6234621  0.56
##               scale(tot_dens_s)  3.738e-02 0.1933384 -0.47 -0.71
## plot:gr:loc:site (Intercept)    2.788e-01 0.5280003
## gr:loc:site      (Intercept)    2.057e-01 0.4535170
```

```

## loc:site      (Intercept)      4.782e-01 0.6915250
## site         (Intercept)      2.188e-08 0.0001479
## Number of obs: 968, groups:
## species, 26; plot:gr:loc:site, 474; gr:loc:site, 110; loc:site, 37; site, 21
##
## Conditional model:
##
##              Estimate Std. Error z value
## (Intercept)      0.66723    0.27771   2.403
## slope.degrees_s  -0.18268    0.07133  -2.561
## scale(tot_dens_s) -0.10921    0.15595  -0.700
## scale(con_dens_s) -0.44496    0.18820  -2.364
## scale(fragment.size) -0.25279    0.15514  -1.629
## trt_II            0.55988    0.15137   3.699
## trt_FF            0.27661    0.13886   1.992
## trt_II:trt_FF     -0.20355    0.21738  -0.936
## scale(tot_dens_s):trt_II 0.04971    0.19245   0.258
## scale(tot_dens_s):trt_FF -0.01093    0.14816  -0.074
## scale(con_dens_s):trt_II -0.25189    0.16952  -1.486
## scale(con_dens_s):trt_FF 0.38152    0.15970   2.389
## scale(tot_dens_s):scale(fragment.size) 0.09861    0.11221   0.879
## scale(con_dens_s):scale(fragment.size) -0.24883    0.16012  -1.554
## scale(fragment.size):trt_II 0.08701    0.12608   0.690
## scale(fragment.size):trt_FF 0.11108    0.11225   0.990
## scale(tot_dens_s):scale(fragment.size):trt_II 0.07160    0.22351   0.320
## scale(tot_dens_s):scale(fragment.size):trt_FF -0.17654    0.20314  -0.869
## scale(con_dens_s):scale(fragment.size):trt_II -0.14412    0.19987  -0.721
## scale(con_dens_s):scale(fragment.size):trt_FF 0.55367    0.19600   2.825
##
##              Pr(>|z|)
## (Intercept)      0.016277 *
## slope.degrees_s  0.010437 *
## scale(tot_dens_s) 0.483771
## scale(con_dens_s) 0.018061 *
## scale(fragment.size) 0.103231
## trt_II            0.000217 ***
## trt_FF            0.046363 *
## trt_II:trt_FF     0.349094
## scale(tot_dens_s):trt_II 0.796159
## scale(tot_dens_s):trt_FF 0.941199
## scale(con_dens_s):trt_II 0.137312
## scale(con_dens_s):trt_FF 0.016894 *
## scale(tot_dens_s):scale(fragment.size) 0.379537
## scale(con_dens_s):scale(fragment.size) 0.120172
## scale(fragment.size):trt_II 0.490132
## scale(fragment.size):trt_FF 0.322352
## scale(tot_dens_s):scale(fragment.size):trt_II 0.748722
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.384811
## scale(con_dens_s):scale(fragment.size):trt_II 0.470860
## scale(con_dens_s):scale(fragment.size):trt_FF 0.004731 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $SR
## Family: binomial ( logit )
## Formula:

```



```

## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##   scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##   scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##   (scale(con_dens_s) + scale(tot_dens_s) | species) + (1 |
##   site/loc/gr/plot)
## Data: filter(sdls, !species == i)
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
##  1538.8   1679.2   -739.4   1478.8     766
##
## Random effects:
##
## Conditional model:
##   Groups          Name              Variance Std.Dev. Corr
##   species          (Intercept)       9.948e-01 0.997383
##                   scale(con_dens_s)  1.666e-01 0.408201  0.49
##                   scale(tot_dens_s)  5.029e-02 0.224248 -0.68 -0.95
##   plot:gr:loc:site (Intercept)       9.819e-02 0.313348
##   gr:loc:site      (Intercept)       1.978e-01 0.444692
##   loc:site         (Intercept)       4.018e-01 0.633879
##   site             (Intercept)       1.277e-08 0.000113
## Number of obs: 796, groups:
## species, 25; plot:gr:loc:site, 436; gr:loc:site, 110; loc:site, 37; site, 21
##
## Conditional model:
##                                     Estimate Std. Error z value
## (Intercept)                        0.63850    0.27529   2.319
## slope.degrees_s                    -0.02056    0.08536  -0.241
## scale(tot_dens_s)                  -0.17944    0.16530  -1.085
## scale(con_dens_s)                  -0.30396    0.15773  -1.927
## scale(fragment.size)               -0.18107    0.15650  -1.157
## trt_II                             0.49663    0.16145   3.076
## trt_FF                             0.41105    0.16944   2.426
## trt_II:trt_FF                     -0.25253    0.24472  -1.032
## scale(tot_dens_s):trt_II           -0.08794    0.22453  -0.392
## scale(tot_dens_s):trt_FF           -0.03447    0.19921  -0.173
## scale(con_dens_s):trt_II           0.02762    0.18700   0.148
## scale(con_dens_s):trt_FF           0.14500    0.18050   0.803
## scale(tot_dens_s):scale(fragment.size) -0.06969    0.14652  -0.476
## scale(con_dens_s):scale(fragment.size) -0.14534    0.15883  -0.915
## scale(fragment.size):trt_II        0.21404    0.14875   1.439
## scale(fragment.size):trt_FF        0.18394    0.15205   1.210
## scale(tot_dens_s):scale(fragment.size):trt_II 0.11592    0.27765   0.418
## scale(tot_dens_s):scale(fragment.size):trt_FF -0.10503    0.25202  -0.417
## scale(con_dens_s):scale(fragment.size):trt_II -0.04202    0.23315  -0.180
## scale(con_dens_s):scale(fragment.size):trt_FF 0.08224    0.23157   0.355
##                                     Pr(>|z|)
## (Intercept)                        0.0204 *
## slope.degrees_s                    0.8096
## scale(tot_dens_s)                  0.2777
## scale(con_dens_s)                  0.0540 .
## scale(fragment.size)              0.2473
## trt_II                            0.0021 **

```

```

## trt_FF                                0.0153 *
## trt_II:trt_FF                        0.3021
## scale(tot_dens_s):trt_II              0.6953
## scale(tot_dens_s):trt_FF              0.8626
## scale(con_dens_s):trt_II              0.8826
## scale(con_dens_s):trt_FF              0.4218
## scale(tot_dens_s):scale(fragment.size) 0.6343
## scale(con_dens_s):scale(fragment.size) 0.3601
## scale(fragment.size):trt_II           0.1502
## scale(fragment.size):trt_FF           0.2264
## scale(tot_dens_s):scale(fragment.size):trt_II 0.6763
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.6769
## scale(con_dens_s):scale(fragment.size):trt_II 0.8570
## scale(con_dens_s):scale(fragment.size):trt_FF 0.7225
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $Symp
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##       scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##       (scale(con_dens_s) + scale(tot_dens_s) | species) + (1 |
##       site/loc/gr/plot)
## Data: filter(sdls, !species == i)
## Weights: census.start
##
##      AIC      BIC    logLik deviance df.resid
##      NA       NA      NA      NA      816
##
## Random effects:
##
## Conditional model:
## Groups      Name                Variance Std.Dev.  Corr
## species      (Intercept)         1.197e+00 1.094e+00
##               scale(con_dens_s)   5.227e-01 7.230e-01 0.47
##               scale(tot_dens_s)   1.641e-02 1.281e-01 -0.36 -0.99
## plot:gr:loc:site (Intercept)     2.593e-01 5.093e-01
## gr:loc:site      (Intercept)     2.719e-01 5.214e-01
## loc:site         (Intercept)     4.625e-01 6.801e-01
## site             (Intercept)     1.189e-295 3.449e-148
## Number of obs: 846, groups:
## species, 25; plot:gr:loc:site, 443; gr:loc:site, 110; loc:site, 37; site, 21
##
## Conditional model:
##
##               Estimate Std. Error z value
## (Intercept)    0.623087  0.299655  2.079
## slope.degrees_s -0.238159  0.078978 -3.016
## scale(tot_dens_s) -0.172296  0.157359 -1.095
## scale(con_dens_s) -0.431181  0.217797 -1.980
## scale(fragment.size) -0.204384  0.161807 -1.263
## trt_II          0.642882  0.164703  3.903
## trt_FF          0.288345  0.145095  1.987

```

```

## trt_II:trt_FF -0.156364 0.231996 -0.674
## scale(tot_dens_s):trt_II -0.045501 0.201656 -0.226
## scale(tot_dens_s):trt_FF -0.003621 0.155956 -0.023
## scale(con_dens_s):trt_II -0.254715 0.190675 -1.336
## scale(con_dens_s):trt_FF 0.465605 0.178066 2.615
## scale(tot_dens_s):scale(fragment.size) 0.098487 0.117733 0.837
## scale(con_dens_s):scale(fragment.size) -0.211136 0.172688 -1.223
## scale(fragment.size):trt_II -0.065994 0.135574 -0.487
## scale(fragment.size):trt_FF 0.090493 0.119279 0.759
## scale(tot_dens_s):scale(fragment.size):trt_II 0.108954 0.228006 0.478
## scale(tot_dens_s):scale(fragment.size):trt_FF -0.251520 0.211014 -1.192
## scale(con_dens_s):scale(fragment.size):trt_II -0.171561 0.220288 -0.779
## scale(con_dens_s):scale(fragment.size):trt_FF 0.650795 0.214434 3.035
## Pr(>|z|)
## (Intercept) 0.03759 *
## slope.degrees_s 0.00257 **
## scale(tot_dens_s) 0.27355
## scale(con_dens_s) 0.04773 *
## scale(fragment.size) 0.20654
## trt_II 9.49e-05 ***
## trt_FF 0.04689 *
## trt_II:trt_FF 0.50032
## scale(tot_dens_s):trt_II 0.82149
## scale(tot_dens_s):trt_FF 0.98148
## scale(con_dens_s):trt_II 0.18160
## scale(con_dens_s):trt_FF 0.00893 **
## scale(tot_dens_s):scale(fragment.size) 0.40286
## scale(con_dens_s):scale(fragment.size) 0.22146
## scale(fragment.size):trt_II 0.62642
## scale(fragment.size):trt_FF 0.44805
## scale(tot_dens_s):scale(fragment.size):trt_II 0.63275
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.23328
## scale(con_dens_s):scale(fragment.size):trt_II 0.43610
## scale(con_dens_s):scale(fragment.size):trt_FF 0.00241 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $Climber1
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
## scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
## scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
## (scale(con_dens_s) + scale(tot_dens_s) | species) + (1 |
## site/loc/gr/plot)
## Data: filter(sdls, !species == i)
## Weights: census.start
##
## AIC BIC logLik deviance df.resid
## 2004.3 2148.1 -972.1 1944.3 861
##
## Random effects:
##
## Conditional model:

```

```

## Groups          Name          Variance Std.Dev.  Corr
## species         (Intercept)    1.110e+00 1.0536851
##                 scale(con_dens_s) 4.279e-01 0.6541436 0.58
##                 scale(tot_dens_s) 3.649e-02 0.1910162 -0.54 -0.62
## plot:gr:loc:site (Intercept)    3.125e-01 0.5589869
## gr:loc:site      (Intercept)    2.224e-01 0.4716092
## loc:site         (Intercept)    4.840e-01 0.6957357
## site             (Intercept)    1.601e-08 0.0001265
## Number of obs: 891, groups:
## species, 25; plot:gr:loc:site, 468; gr:loc:site, 110; loc:site, 37; site, 21
##
## Conditional model:
##
##                                     Estimate Std. Error z value
## (Intercept)                        0.72418    0.29263    2.475
## slope.degrees_s                    -0.18938    0.07488   -2.529
## scale(tot_dens_s)                  -0.05951    0.17064   -0.349
## scale(con_dens_s)                  -0.49321    0.21218   -2.325
## scale(fragment.size)               -0.27528    0.16120   -1.708
## trt_II                             0.52847    0.16187    3.265
## trt_FF                             0.33985    0.14975    2.269
## trt_II:trt_FF                     -0.23080    0.23385   -0.987
## scale(tot_dens_s):trt_II           0.02794    0.19797    0.141
## scale(tot_dens_s):trt_FF           -0.02253    0.15547   -0.145
## scale(con_dens_s):trt_II           -0.24455    0.19155   -1.277
## scale(con_dens_s):trt_FF           0.42565    0.18361    2.318
## scale(tot_dens_s):scale(fragment.size) 0.13497    0.11919    1.132
## scale(con_dens_s):scale(fragment.size) -0.25574    0.17360   -1.473
## scale(fragment.size):trt_II        0.05831    0.13213    0.441
## scale(fragment.size):trt_FF        0.10946    0.11909    0.919
## scale(tot_dens_s):scale(fragment.size):trt_II 0.01717    0.23536    0.073
## scale(tot_dens_s):scale(fragment.size):trt_FF -0.16414    0.22013   -0.746
## scale(con_dens_s):scale(fragment.size):trt_II -0.10702    0.22543   -0.475
## scale(con_dens_s):scale(fragment.size):trt_FF 0.57845    0.22089    2.619
##                                     Pr(>|z|)
## (Intercept)                        0.01333 *
## slope.degrees_s                    0.01144 *
## scale(tot_dens_s)                  0.72728
## scale(con_dens_s)                  0.02010 *
## scale(fragment.size)              0.08770 .
## trt_II                             0.00110 **
## trt_FF                             0.02324 *
## trt_II:trt_FF                     0.32367
## scale(tot_dens_s):trt_II           0.88777
## scale(tot_dens_s):trt_FF           0.88479
## scale(con_dens_s):trt_II           0.20171
## scale(con_dens_s):trt_FF           0.02044 *
## scale(tot_dens_s):scale(fragment.size) 0.25748
## scale(con_dens_s):scale(fragment.size) 0.14071
## scale(fragment.size):trt_II        0.65901
## scale(fragment.size):trt_FF        0.35802
## scale(tot_dens_s):scale(fragment.size):trt_II 0.94184
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.45590
## scale(con_dens_s):scale(fragment.size):trt_II 0.63498
## scale(con_dens_s):scale(fragment.size):trt_FF 0.00882 **

```

```

## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $FLC
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##       scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##       (scale(con_dens_s) + scale(tot_dens_s) | species) + (1 |
##       site/loc/gr/plot)
## Data: filter(sdls, !species == i)
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
## 2081.8   2226.0 -1010.9   2021.8      874
##
## Random effects:
##
## Conditional model:
## Groups          Name              Variance Std.Dev.  Corr
## species          (Intercept)      1.054e+00 1.0267284
##                  scale(con_dens_s) 5.047e-01 0.7104215  0.55
##                  scale(tot_dens_s) 8.428e-02 0.2903098 -0.44 -0.72
## plot:gr:loc:site (Intercept)      2.697e-01 0.5193591
## gr:loc:site      (Intercept)      1.919e-01 0.4380878
## loc:site         (Intercept)      4.780e-01 0.6913691
## site             (Intercept)      2.726e-08 0.0001651
## Number of obs: 904, groups:
## species, 25; plot:gr:loc:site, 466; gr:loc:site, 110; loc:site, 37; site, 21
##
## Conditional model:
##
##              Estimate Std. Error z value
## (Intercept)      0.60130    0.28614   2.101
## slope.degrees_s  -0.19046    0.07263  -2.622
## scale(tot_dens_s) -0.24177    0.17930  -1.348
## scale(con_dens_s) -0.35898    0.21142  -1.698
## scale(fragment.size) -0.26189    0.15705  -1.668
## trt_II            0.54049    0.15528   3.481
## trt_FF            0.27584    0.14241   1.937
## trt_II:trt_FF     -0.19031    0.22314  -0.853
## scale(tot_dens_s):trt_II  0.10583    0.20414   0.518
## scale(tot_dens_s):trt_FF  0.04725    0.15841   0.298
## scale(con_dens_s):trt_II -0.25219    0.18033  -1.398
## scale(con_dens_s):trt_FF  0.35289    0.16866   2.092
## scale(tot_dens_s):scale(fragment.size) 0.17530    0.11231   1.561
## scale(con_dens_s):scale(fragment.size) -0.28594    0.17139  -1.668
## scale(fragment.size):trt_II  0.04834    0.12808   0.377
## scale(fragment.size):trt_FF  0.16345    0.11585   1.411
## scale(tot_dens_s):scale(fragment.size):trt_II -0.05304    0.23021  -0.230
## scale(tot_dens_s):scale(fragment.size):trt_FF -0.13798    0.21737  -0.635
## scale(con_dens_s):scale(fragment.size):trt_II -0.07390    0.21099  -0.350
## scale(con_dens_s):scale(fragment.size):trt_FF  0.51571    0.20874   2.471
##
##              Pr(>|z|)

```

```

## (Intercept) 0.03560 *
## slope.degrees_s 0.00873 **
## scale(tot_dens_s) 0.17753
## scale(con_dens_s) 0.08951 .
## scale(fragment.size) 0.09540 .
## trt_II 0.00050 ***
## trt_FF 0.05274 .
## trt_II:trt_FF 0.39374
## scale(tot_dens_s):trt_II 0.60416
## scale(tot_dens_s):trt_FF 0.76549
## scale(con_dens_s):trt_II 0.16196
## scale(con_dens_s):trt_FF 0.03641 *
## scale(tot_dens_s):scale(fragment.size) 0.11854
## scale(con_dens_s):scale(fragment.size) 0.09525 .
## scale(fragment.size):trt_II 0.70588
## scale(fragment.size):trt_FF 0.15828
## scale(tot_dens_s):scale(fragment.size):trt_II 0.81778
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.52559
## scale(con_dens_s):scale(fragment.size):trt_II 0.72616
## scale(con_dens_s):scale(fragment.size):trt_FF 0.01349 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $Cinam
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##       scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##       (scale(con_dens_s) + scale(tot_dens_s) | species) + (1 |
##       site/loc/gr/plot)
## Data: filter(sdls, !species == i)
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
## 2059.4  2202.9  -999.7  1999.4      852
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.  Corr
## species      (Intercept)  9.639e-01 0.9817926
##              scale(con_dens_s) 4.032e-01 0.6349760  0.45
##              scale(tot_dens_s) 4.948e-02 0.2224359 -0.38 -0.79
## plot:gr:loc:site (Intercept)  2.906e-01 0.5390979
## gr:loc:site      (Intercept)  2.685e-01 0.5181503
## loc:site         (Intercept)  4.422e-01 0.6649552
## site            (Intercept)  2.109e-08 0.0001452
## Number of obs: 882, groups:
## species, 25; plot:gr:loc:site, 461; gr:loc:site, 110; loc:site, 37; site, 21
##
## Conditional model:
##
## Estimate Std. Error z value
## (Intercept) 0.70050 0.27933 2.508

```

```
## slope.degrees_s -0.19170 0.07382 -2.597
## scale(tot_dens_s) -0.13076 0.16765 -0.780
## scale(con_dens_s) -0.48358 0.19780 -2.445
## scale(fragment.size) -0.27909 0.15808 -1.765
## trt_II 0.59369 0.15775 3.763
## trt_FF 0.31667 0.14362 2.205
## trt_II:trt_FF -0.24942 0.22545 -1.106
## scale(tot_dens_s):trt_II 0.05476 0.20594 0.266
## scale(tot_dens_s):trt_FF -0.03671 0.15834 -0.232
## scale(con_dens_s):trt_II -0.27294 0.18369 -1.486
## scale(con_dens_s):trt_FF 0.47156 0.17190 2.743
## scale(tot_dens_s):scale(fragment.size) 0.11020 0.12150 0.907
## scale(con_dens_s):scale(fragment.size) -0.35449 0.16559 -2.141
## scale(fragment.size):trt_II 0.09773 0.13213 0.740
## scale(fragment.size):trt_FF 0.10274 0.11743 0.875
## scale(tot_dens_s):scale(fragment.size):trt_II 0.09621 0.23656 0.407
## scale(tot_dens_s):scale(fragment.size):trt_FF -0.23337 0.21662 -1.077
## scale(con_dens_s):scale(fragment.size):trt_II -0.17272 0.21472 -0.804
## scale(con_dens_s):scale(fragment.size):trt_FF 0.64615 0.21256 3.040
## Pr(>|z|)
## (Intercept) 0.012149 *
## slope.degrees_s 0.009406 **
## scale(tot_dens_s) 0.435430
## scale(con_dens_s) 0.014492 *
## scale(fragment.size) 0.077487 .
## trt_II 0.000168 ***
## trt_FF 0.027460 *
## trt_II:trt_FF 0.268580
## scale(tot_dens_s):trt_II 0.790319
## scale(tot_dens_s):trt_FF 0.816674
## scale(con_dens_s):trt_II 0.137320
## scale(con_dens_s):trt_FF 0.006083 **
## scale(tot_dens_s):scale(fragment.size) 0.364392
## scale(con_dens_s):scale(fragment.size) 0.032297 *
## scale(fragment.size):trt_II 0.459497
## scale(fragment.size):trt_FF 0.381631
## scale(tot_dens_s):scale(fragment.size):trt_II 0.684220
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.281345
## scale(con_dens_s):scale(fragment.size):trt_II 0.421187
## scale(con_dens_s):scale(fragment.size):trt_FF 0.002366 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Only species that makes a difference is *S. rubicundum* - removing it dampens interaction between density and fungicide (it is still there, but marginally non-significant)

what about single species models?

```
## remove species random effect
names(sp_common) <- sp_common
single_sp_mods <- map(sp_common[1:5], function(i) {
  glmmTMB(Pr_s ~ slope.degrees_s +
    (scale(tot_dens_s) + scale(con_dens_s)) +
    scale(fragment.size) +
```

```

      trt_I*trt_F +
      (scale(tot_dens_s) + scale(con_dens_s)) *
      (trt_I + trt_F) *
      scale(fragment.size) +
      ## setting cor to 0 to converge
      (1|site/loc/gr/plot),
weights = census.start,
data = filter(sdls, species == i),
family=binomial)})

names(single_sp_mods) <-
  sp_codes$spbin[match(names(single_sp_mods), sp_codes$code)]
single_sp_mods$All <- m_cdd_s_ris

map(single_sp_mods, summary)

## $'Syzygium rubicundum'
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##       scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##       (1 | site/loc/gr/plot)
## Data: filter(sdls, species == i)
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
##    664.8    740.4   -308.4    616.8     148
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## plot:gr:loc:site (Intercept) 2.103e-01 0.4586119
## gr:loc:site      (Intercept) 1.693e-01 0.4114793
## loc:site         (Intercept) 6.510e-01 0.8068493
## site             (Intercept) 1.922e-07 0.0004384
## Number of obs: 172, groups:
## plot:gr:loc:site, 172; gr:loc:site, 61; loc:site, 30; site, 16
##
## Conditional model:
##
##              Estimate Std. Error z value
## (Intercept)   -0.14989    0.23065  -0.650
## slope.degrees_s -0.31491    0.10682  -2.948
## scale(tot_dens_s)  0.04320    0.14091   0.307
## scale(con_dens_s) -0.65441    0.16022  -4.084
## scale(fragment.size) -0.19506    0.23153  -0.842
## trt_II          0.69260    0.25888   2.675
## trt_FF          -0.01042    0.20949  -0.050
## trt_II:trt_FF    -0.24727    0.33821  -0.731
## scale(tot_dens_s):trt_II -0.04539    0.34754  -0.131
## scale(tot_dens_s):trt_FF -0.05204    0.18634  -0.279
## scale(con_dens_s):trt_II  0.06969    0.24165   0.288

```



```

## scale(con_dens_s):trt_FF          0.07628    0.21351    0.357
## scale(tot_dens_s):scale(fragment.size) 0.32513    0.12538    2.593
## scale(con_dens_s):scale(fragment.size) -0.08381    0.11571   -0.724
## scale(fragment.size):trt_II        -0.07710    0.24413   -0.316
## scale(fragment.size):trt_FF        -0.28855    0.20967   -1.376
## scale(tot_dens_s):scale(fragment.size):trt_II -0.42050    0.33640   -1.250
## scale(tot_dens_s):scale(fragment.size):trt_FF -0.26047    0.33499   -0.778
## scale(con_dens_s):scale(fragment.size):trt_II -0.01368    0.17338   -0.079
## scale(con_dens_s):scale(fragment.size):trt_FF  0.37473    0.15094    2.483
##                                     Pr(>|z|)
## (Intercept)                        0.51578
## slope.degrees_s                    0.00320 **
## scale(tot_dens_s)                   0.75919
## scale(con_dens_s)                   4.42e-05 ***
## scale(fragment.size)                0.39952
## trt_II                             0.00746 **
## trt_FF                             0.96033
## trt_II:trt_FF                      0.46470
## scale(tot_dens_s):trt_II            0.89608
## scale(tot_dens_s):trt_FF            0.78005
## scale(con_dens_s):trt_II            0.77306
## scale(con_dens_s):trt_FF            0.72090
## scale(tot_dens_s):scale(fragment.size) 0.00951 **
## scale(con_dens_s):scale(fragment.size) 0.46888
## scale(fragment.size):trt_II         0.75214
## scale(fragment.size):trt_FF         0.16877
## scale(tot_dens_s):scale(fragment.size):trt_II 0.21130
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.43684
## scale(con_dens_s):scale(fragment.size):trt_II 0.93710
## scale(con_dens_s):scale(fragment.size):trt_FF 0.01304 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $'Symplocos racemosa'
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##       scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##       (1 | site/loc/gr/plot)
## Data: filter(sdls, species == i)
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
##    328.1    395.4   -140.0    280.1      98
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## plot:gr:loc:site (Intercept) 1.625e-01 0.4030853
## gr:loc:site      (Intercept) 2.503e-02 0.1582100
## loc:site         (Intercept) 5.766e-01 0.7593371
## site             (Intercept) 1.487e-08 0.0001219

```

```

## Number of obs: 122, groups:
## plot:gr:loc:site, 122; gr:loc:site, 53; loc:site, 24; site, 17
##
## Conditional model:
##
##               Estimate Std. Error z value
## (Intercept)      0.71220    0.34523   2.063
## slope.degrees_s    0.19652    0.15944   1.233
## scale(tot_dens_s)   0.30384    0.33337   0.911
## scale(con_dens_s)   0.25698    0.61914   0.415
## scale(fragment.size) -0.22861    0.43806  -0.522
## trt_II             -0.26647    0.48631  -0.548
## trt_FF              1.60922    0.79724   2.018
## trt_II:trt_FF      -1.12097    0.69201  -1.620
## scale(tot_dens_s):trt_II -0.04651    0.73245  -0.064
## scale(tot_dens_s):trt_FF  1.53285    1.47131   1.042
## scale(con_dens_s):trt_II -1.73483    1.47049  -1.180
## scale(con_dens_s):trt_FF  0.89298    2.64710   0.337
## scale(tot_dens_s):scale(fragment.size) 0.10147    0.39163   0.259
## scale(con_dens_s):scale(fragment.size) 0.69530    0.85389   0.814
## scale(fragment.size):trt_II -0.33443    0.63470  -0.527
## scale(fragment.size):trt_FF  1.42770    0.91043   1.568
## scale(tot_dens_s):scale(fragment.size):trt_II -0.79311    0.95984  -0.826
## scale(tot_dens_s):scale(fragment.size):trt_FF  2.43647    2.06836   1.178
## scale(con_dens_s):scale(fragment.size):trt_II -2.56759    1.99589  -1.286
## scale(con_dens_s):scale(fragment.size):trt_FF  1.12732    3.66319   0.308
##
##               Pr(>|z|)
## (Intercept)      0.0391 *
## slope.degrees_s    0.2177
## scale(tot_dens_s)   0.3621
## scale(con_dens_s)   0.6781
## scale(fragment.size) 0.6018
## trt_II             0.5837
## trt_FF              0.0435 *
## trt_II:trt_FF      0.1053
## scale(tot_dens_s):trt_II 0.9494
## scale(tot_dens_s):trt_FF 0.2975
## scale(con_dens_s):trt_II 0.2381
## scale(con_dens_s):trt_FF 0.7359
## scale(tot_dens_s):scale(fragment.size) 0.7956
## scale(con_dens_s):scale(fragment.size) 0.4155
## scale(fragment.size):trt_II 0.5983
## scale(fragment.size):trt_FF 0.1168
## scale(tot_dens_s):scale(fragment.size):trt_II 0.4086
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.2388
## scale(con_dens_s):scale(fragment.size):trt_II 0.1983
## scale(con_dens_s):scale(fragment.size):trt_FF 0.7583
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $'Spatholobus purpureus'
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +

```

```

##      scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##      (1 | site/loc/gr/plot)
## Data: filter(sdls, species == i)
## Weights: census.start
##
##      AIC      BIC    logLik deviance df.resid
##      221.1    277.4    -86.6    173.1      53
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## plot:gr:loc:site (Intercept) 3.501e-11 5.917e-06
## gr:loc:site      (Intercept) 6.978e-12 2.642e-06
## loc:site         (Intercept) 1.120e-10 1.058e-05
## site             (Intercept) 3.560e-11 5.967e-06
## Number of obs: 77, groups:
## plot:gr:loc:site, 77; gr:loc:site, 30; loc:site, 15; site, 10
##
## Conditional model:
##                                     Estimate Std. Error z value
## (Intercept)                       -0.88691    0.24982  -3.550
## slope.degrees_s                   -0.40153    0.12585  -3.191
## scale(tot_dens_s)                 -1.01314    0.52171  -1.942
## scale(con_dens_s)                 -0.03815    0.13710  -0.278
## scale(fragment.size)              -0.36087    0.24130  -1.496
## trt_II                            1.22403    0.48781   2.509
## trt_FF                            0.41325    0.37907   1.090
## trt_II:trt_FF                    -0.31901    0.42278  -0.755
## scale(tot_dens_s):trt_II          0.17701    0.88399   0.200
## scale(tot_dens_s):trt_FF          0.78859    0.65603   1.202
## scale(con_dens_s):trt_II         -0.20655    0.26658  -0.775
## scale(con_dens_s):trt_FF         -0.18489    0.22084  -0.837
## scale(tot_dens_s):scale(fragment.size) -0.77759    0.50864  -1.529
## scale(con_dens_s):scale(fragment.size) -1.55347    0.86887  -1.788
## scale(fragment.size):trt_II      -1.70031    1.44116  -1.180
## scale(fragment.size):trt_FF       1.48540    0.63381   2.344
## scale(tot_dens_s):scale(fragment.size):trt_II 0.42592    0.79637   0.535
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.40897    0.63428   0.645
## scale(con_dens_s):scale(fragment.size):trt_II -5.94422    4.07890  -1.457
## scale(con_dens_s):scale(fragment.size):trt_FF 3.21763    1.72268   1.868
##                                     Pr(>|z|)
## (Intercept)                       0.000385 ***
## slope.degrees_s                   0.001420 **
## scale(tot_dens_s)                 0.052141 .
## scale(con_dens_s)                 0.780831
## scale(fragment.size)              0.134775
## trt_II                            0.012099 *
## trt_FF                            0.275639
## trt_II:trt_FF                    0.450521
## scale(tot_dens_s):trt_II          0.841293
## scale(tot_dens_s):trt_FF          0.229340
## scale(con_dens_s):trt_II          0.438433
## scale(con_dens_s):trt_FF          0.402457

```

```

## scale(tot_dens_s):scale(fragment.size)      0.126322
## scale(con_dens_s):scale(fragment.size)      0.073789 .
## scale(fragment.size):trt_II                 0.238070
## scale(fragment.size):trt_FF                 0.019098 *
## scale(tot_dens_s):scale(fragment.size):trt_II 0.592770
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.519066
## scale(con_dens_s):scale(fragment.size):trt_II 0.145030
## scale(con_dens_s):scale(fragment.size):trt_FF 0.061790 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $'Ventilago madraspatana'
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##       scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##       (1 | site/loc/gr/plot)
## Data: filter(sdls, species == i)
## Weights: census.start
##
##      AIC      BIC    logLik deviance df.resid
##    140.5    192.4    -46.3     92.5      40
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## plot:gr:loc:site (Intercept) 1.595e-10 1.263e-05
## gr:loc:site      (Intercept) 3.794e-17 6.159e-09
## loc:site         (Intercept) 3.726e-13 6.104e-07
## site             (Intercept) 2.325e-13 4.821e-07
## Number of obs: 64, groups:
## plot:gr:loc:site, 64; gr:loc:site, 27; loc:site, 16; site, 10
##
## Conditional model:
##
##              Estimate Std. Error z value
## (Intercept)      1.08195    0.43536   2.485
## slope.degrees_s  -0.33295    0.25156  -1.324
## scale(tot_dens_s)  0.77152    0.99666   0.774
## scale(con_dens_s) -1.10041    0.66212  -1.662
## scale(fragment.size) 0.32157    0.41979   0.766
## trt_II            5.03424    2.00733   2.508
## trt_FF            0.01163    0.61168   0.019
## trt_II:trt_FF     -2.26606    1.09780  -2.064
## scale(tot_dens_s):trt_II -3.30626    3.59749  -0.919
## scale(tot_dens_s):trt_FF -0.16767    0.93832  -0.179
## scale(con_dens_s):trt_II -7.36897    3.06914  -2.401
## scale(con_dens_s):trt_FF  1.17060    1.27183   0.920
## scale(tot_dens_s):scale(fragment.size) -1.60292    0.90233  -1.776
## scale(con_dens_s):scale(fragment.size)  1.32854    1.18201   1.124
## scale(fragment.size):trt_II  4.12978    2.49481   1.655
## scale(fragment.size):trt_FF -1.07212    0.74249  -1.444
## scale(tot_dens_s):scale(fragment.size):trt_II -3.00156    5.18441  -0.579

```

```

## scale(tot_dens_s):scale(fragment.size):trt_FF  0.20736    1.29481    0.160
## scale(con_dens_s):scale(fragment.size):trt_II -10.58223    4.14115   -2.555
## scale(con_dens_s):scale(fragment.size):trt_FF   0.54595    1.81312    0.301
##                                     Pr(>|z|)
## (Intercept)                                0.0129 *
## slope.degrees_s                            0.1856
## scale(tot_dens_s)                           0.4389
## scale(con_dens_s)                           0.0965 .
## scale(fragment.size)                        0.4437
## trt_II                                       0.0121 *
## trt_FF                                       0.9848
## trt_II:trt_FF                               0.0390 *
## scale(tot_dens_s):trt_II                    0.3581
## scale(tot_dens_s):trt_FF                    0.8582
## scale(con_dens_s):trt_II                    0.0164 *
## scale(con_dens_s):trt_FF                    0.3574
## scale(tot_dens_s):scale(fragment.size)      0.0757 .
## scale(con_dens_s):scale(fragment.size)      0.2610
## scale(fragment.size):trt_II                 0.0979 .
## scale(fragment.size):trt_FF                 0.1488
## scale(tot_dens_s):scale(fragment.size):trt_II 0.5626
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.8728
## scale(con_dens_s):scale(fragment.size):trt_II 0.0106 *
## scale(con_dens_s):scale(fragment.size):trt_FF 0.7633
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $'Cinnamomum sp.'
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##       scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##       (1 | site/loc/gr/plot)
## Data: filter(sdls, species == i)
## Weights: census.start
##
##      AIC      BIC   logLik deviance df.resid
##    158.7    217.6    -55.3    110.7      62
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## plot:gr:loc:site (Intercept) 2.994e-09 5.472e-05
## gr:loc:site      (Intercept) 9.293e-01 9.640e-01
## loc:site         (Intercept) 3.086e-09 5.555e-05
## site             (Intercept) 7.390e-01 8.597e-01
## Number of obs: 86, groups:
## plot:gr:loc:site, 86; gr:loc:site, 42; loc:site, 23; site, 17
##
## Conditional model:
##                                     Estimate Std. Error z value
## (Intercept)                        -0.71081    0.57847  -1.229

```

```

## slope.degrees_s          -0.21513    0.49865   -0.431
## scale(tot_dens_s)        -0.82516    0.59356   -1.390
## scale(con_dens_s)         0.37420    0.50413    0.742
## scale(fragment.size)     -0.40154    0.59974   -0.669
## trt_II                    0.41253    0.82832    0.498
## trt_FF                    1.92466    1.11891    1.720
## trt_II:trt_FF            -1.35961    1.49140   -0.912
## scale(tot_dens_s):trt_II  -0.18009    1.13034   -0.159
## scale(tot_dens_s):trt_FF   3.47128    1.77665    1.954
## scale(con_dens_s):trt_II  -0.38512    1.64299   -0.234
## scale(con_dens_s):trt_FF  -0.33156    1.60517   -0.207
## scale(tot_dens_s):scale(fragment.size)  0.59999    0.40618    1.477
## scale(con_dens_s):scale(fragment.size)  0.36357    0.58390    0.623
## scale(fragment.size):trt_II -1.45627    1.03267   -1.410
## scale(fragment.size):trt_FF   1.92983    1.33244    1.448
## scale(tot_dens_s):scale(fragment.size):trt_II -1.81786    1.65778   -1.097
## scale(tot_dens_s):scale(fragment.size):trt_FF  3.45466    2.25046    1.535
## scale(con_dens_s):scale(fragment.size):trt_II  0.07232    2.04916    0.035
## scale(con_dens_s):scale(fragment.size):trt_FF  0.98437    2.05481    0.479
##                               Pr(>|z|)
## (Intercept)                0.2192
## slope.degrees_s             0.6662
## scale(tot_dens_s)            0.1645
## scale(con_dens_s)            0.4579
## scale(fragment.size)         0.5032
## trt_II                       0.6185
## trt_FF                       0.0854 .
## trt_II:trt_FF                0.3620
## scale(tot_dens_s):trt_II      0.8734
## scale(tot_dens_s):trt_FF      0.0507 .
## scale(con_dens_s):trt_II      0.8147
## scale(con_dens_s):trt_FF      0.8364
## scale(tot_dens_s):scale(fragment.size)  0.1396
## scale(con_dens_s):scale(fragment.size)  0.5335
## scale(fragment.size):trt_II    0.1585
## scale(fragment.size):trt_FF    0.1475
## scale(tot_dens_s):scale(fragment.size):trt_II  0.2728
## scale(tot_dens_s):scale(fragment.size):trt_FF  0.1248
## scale(con_dens_s):scale(fragment.size):trt_II  0.9718
## scale(con_dens_s):scale(fragment.size):trt_FF  0.6319
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $All
## Family: binomial ( logit )
## Formula:
## Pr_s ~ slope.degrees_s + (scale(tot_dens_s) + scale(con_dens_s)) +
##       scale(fragment.size) + trt_I * trt_F + (scale(tot_dens_s) +
##       scale(con_dens_s)) * (trt_I + trt_F) * scale(fragment.size) +
##       (scale(con_dens_s) + scale(tot_dens_s) | species) + (1 |
##       site/loc/gr/plot)
## Data: sdls
## Weights: census.start
##

```

```

##      AIC      BIC    logLik deviance df.resid
##    2215.0    2361.3  -1077.5   2155.0     938
##
## Random effects:
##
## Conditional model:
##   Groups      Name              Variance Std.Dev.  Corr
##   species      (Intercept)        1.027e+00 1.0131846
##               scale(con_dens_s)  3.887e-01 0.6234621  0.56
##               scale(tot_dens_s)  3.738e-02 0.1933384 -0.47 -0.71
##   plot:gr:loc:site (Intercept)    2.788e-01 0.5280003
##   gr:loc:site      (Intercept)    2.057e-01 0.4535170
##   loc:site         (Intercept)    4.782e-01 0.6915250
##   site             (Intercept)    2.188e-08 0.0001479
## Number of obs: 968, groups:
## species, 26; plot:gr:loc:site, 474; gr:loc:site, 110; loc:site, 37; site, 21
##
## Conditional model:
##                                     Estimate Std. Error z value
## (Intercept)                        0.66723    0.27771    2.403
## slope.degrees_s                    -0.18268    0.07133   -2.561
## scale(tot_dens_s)                  -0.10921    0.15595   -0.700
## scale(con_dens_s)                  -0.44496    0.18820   -2.364
## scale(fragment.size)               -0.25279    0.15514   -1.629
## trt_II                             0.55988    0.15137    3.699
## trt_FF                             0.27661    0.13886    1.992
## trt_II:trt_FF                     -0.20355    0.21738   -0.936
## scale(tot_dens_s):trt_II           0.04971    0.19245    0.258
## scale(tot_dens_s):trt_FF          -0.01093    0.14816   -0.074
## scale(con_dens_s):trt_II          -0.25189    0.16952   -1.486
## scale(con_dens_s):trt_FF           0.38152    0.15970    2.389
## scale(tot_dens_s):scale(fragment.size) 0.09861    0.11221    0.879
## scale(con_dens_s):scale(fragment.size) -0.24883    0.16012   -1.554
## scale(fragment.size):trt_II        0.08701    0.12608    0.690
## scale(fragment.size):trt_FF        0.11108    0.11225    0.990
## scale(tot_dens_s):scale(fragment.size):trt_II 0.07160    0.22351    0.320
## scale(tot_dens_s):scale(fragment.size):trt_FF -0.17654    0.20314   -0.869
## scale(con_dens_s):scale(fragment.size):trt_II -0.14412    0.19987   -0.721
## scale(con_dens_s):scale(fragment.size):trt_FF 0.55367    0.19600    2.825
##                                     Pr(>|z|)
## (Intercept)                        0.016277 *
## slope.degrees_s                    0.010437 *
## scale(tot_dens_s)                   0.483771
## scale(con_dens_s)                   0.018061 *
## scale(fragment.size)                0.103231
## trt_II                             0.000217 ***
## trt_FF                             0.046363 *
## trt_II:trt_FF                      0.349094
## scale(tot_dens_s):trt_II            0.796159
## scale(tot_dens_s):trt_FF            0.941199
## scale(con_dens_s):trt_II            0.137312
## scale(con_dens_s):trt_FF            0.016894 *
## scale(tot_dens_s):scale(fragment.size) 0.379537
## scale(con_dens_s):scale(fragment.size) 0.120172

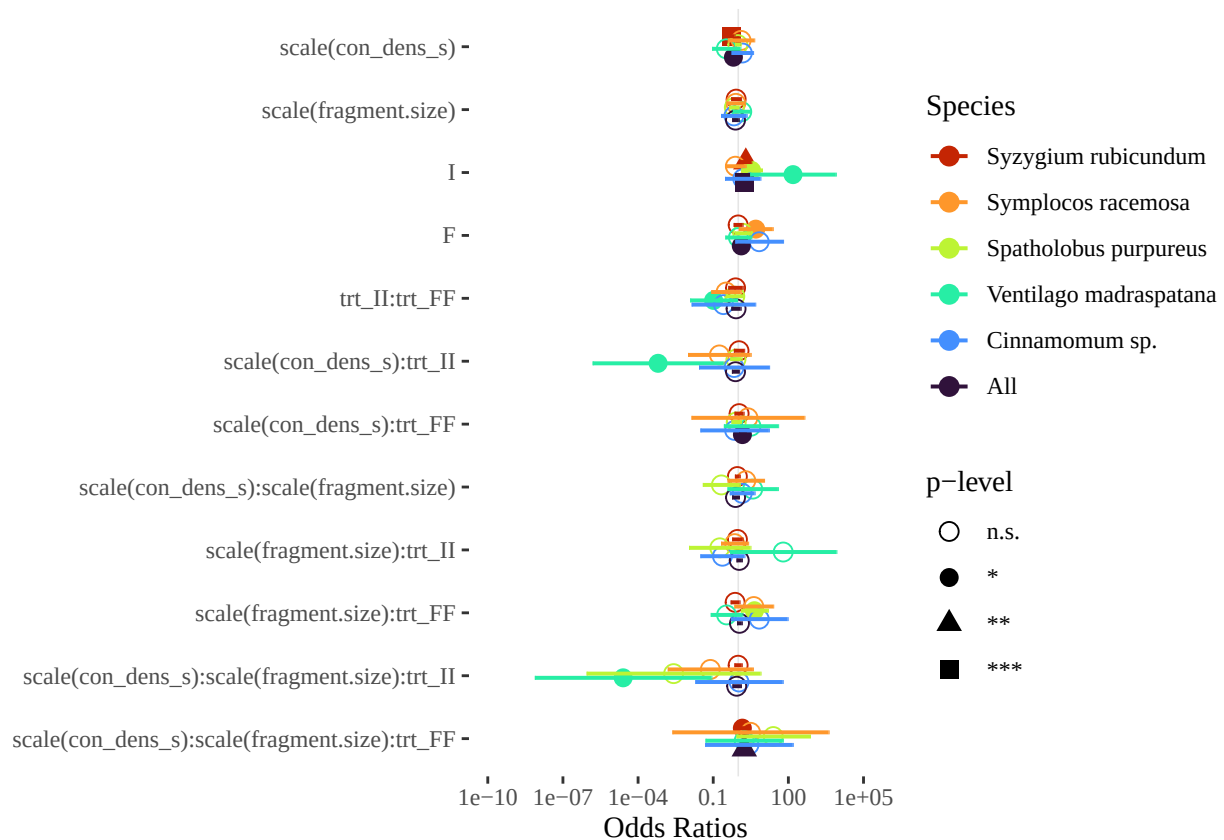
```

```
## scale(fragment.size):trt_II 0.490132
## scale(fragment.size):trt_FF 0.322352
## scale(tot_dens_s):scale(fragment.size):trt_II 0.748722
## scale(tot_dens_s):scale(fragment.size):trt_FF 0.384811
## scale(con_dens_s):scale(fragment.size):trt_II 0.470860
## scale(con_dens_s):scale(fragment.size):trt_FF 0.004731 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
term_nms <- names(fixef(single_sp_mods[[1]])$cond)

plot_models(single_sp_mods, m.labels=names(single_sp_mods), p.shape=TRUE,
            rm.terms = c("slope.degrees_s",
                        term_nms[str_detect(term_nms, "tot")])) +
  scale_colour_viridis_d(end=0.9, name="Species", option="H")
```

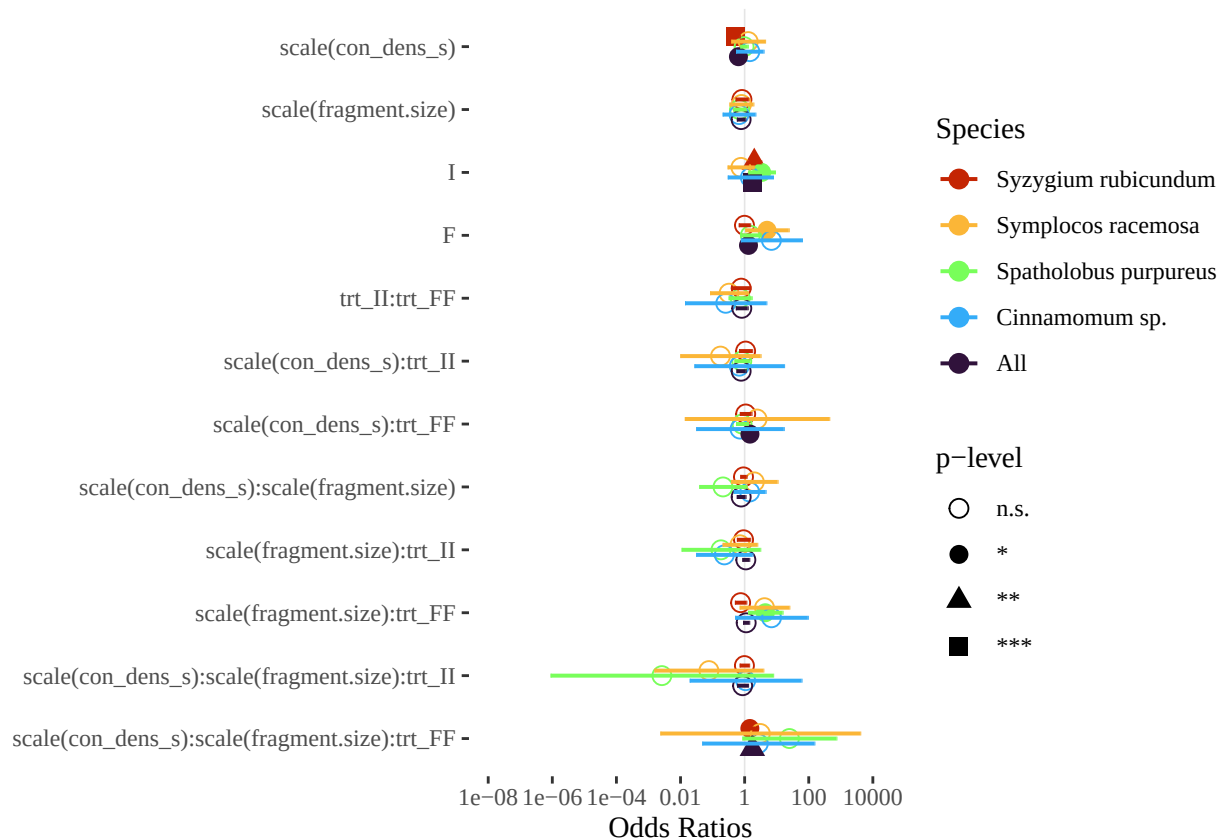
```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



```
## Ventilago distorts scale, so dropping it
plot_models(single_sp_mods[-4], m.labels=names(single_sp_mods)[-4],
            p.shape=TRUE,
            rm.terms = c("slope.degrees_s",
                        term_nms[str_detect(term_nms, "tot")])) +
  scale_colour_viridis_d(end=0.9, name="Species", option="H")
```



```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



```
## As convenient as the plot_models function is, it would be nice to
## plot different species on different scales and simplify the
## p-values
```

```
sp_effects <- map(single_sp_mods, \(d) {
  tidy(d, conf.int = TRUE) |>
  filter(effect == "fixed", str_detect(term, "tot", negate=TRUE),
    !(term %in% c("(Intercept)", "slope.degrees_s")))
}) |> bind_rows(.id = "Species")

labs <- data.frame(par = c("ConDens", "FragArea", "I", "F", "I:F",
  "ConDens:I", "ConDens:F", "FragArea:\n ConDens",
  "FragArea:I", "FragArea:F",
  "FragArea:\n (ConDens:I)", "FragArea:\n (ConDens:F)"),
  term = unique(sp_effects$term)) |>
  mutate(par = factor(par, levels = rev(par)))

sp_effects <- left_join(sp_effects, labs, by = "term")

sp_effects <- mutate(sp_effects,
  Biocide = case_when(
    str_detect(term, "trt_II") ~ "Insecticide",
```

```

      str_detect(term, "trt_FF") ~ "Fungicide",
      .default = "Control"),
  Biocide = factor(
    ifelse(str_detect(term,"trt_II:trt_FF"),
           "Both", Biocide),
    levels = c("Control", "Insecticide",
               "Fungicide", "Both"))
sp_effects <- sp_effects |> mutate(sp_effects,
                                  Species = ifelse(Species == "All", "All",
                                                    paste0("italic('", Species, "')")))

pl_species <-
  filter(sp_effects, Species != "All", par != "I:F") |>
  ggplot(aes(x = estimate, xmin = conf.low, xmax = conf.high,
             y = par, colour = Biocide)) +
    facet_wrap(~Species, scales = "free_x", labeller = label_parsed) +
    geom_pointrange(aes(shape = p.value < 0.05 )) +
    geom_vline(xintercept=0, linetype = "dotted") +
    scale_colour_brewer(palette="Set2") +
    scale_shape_manual(values=c(21, 16), guide = "none" ) +
    labs(y = NULL, x = "log(odds ratio)")

pl_species <- ((blup_plot) |
              (pl_species + theme(legend.position = c(0.95, 0.05),
                                   legend.justification=c(1, 0)))) +
  plot_layout(widths=c(0.35, 0.65)) + plot_annotation(tag_levels = "A")

```

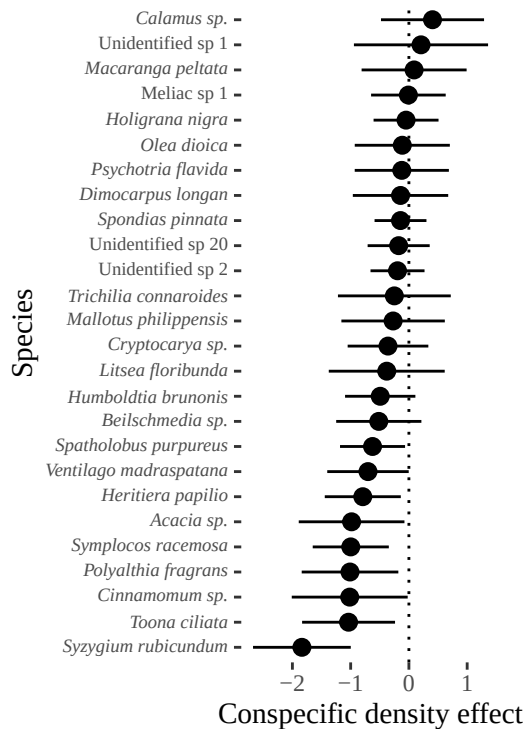
```

## Warning: A numeric 'legend.position' argument in 'theme()' was deprecated in ggplot2
## 3.5.0.
## i Please use the 'legend.position.inside' argument of 'theme()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

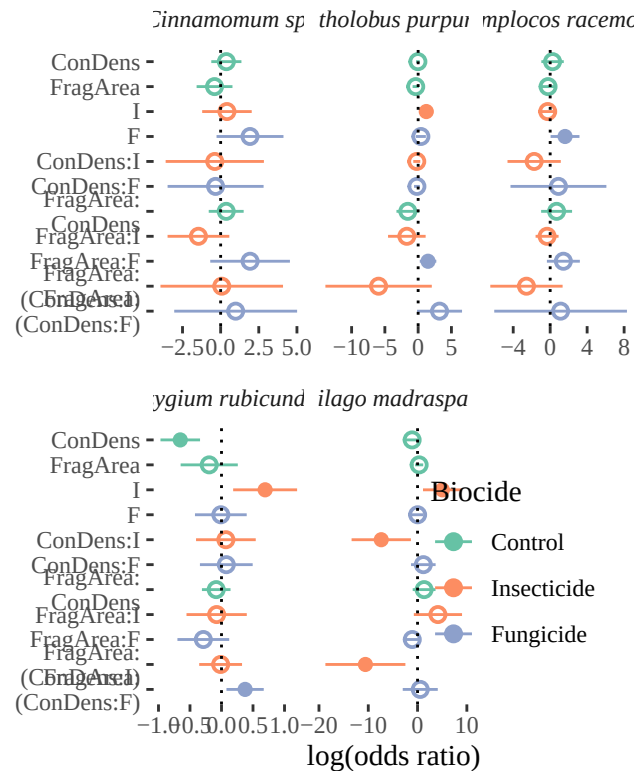
```

```
pl_species
```

A



B



```
ggsave(pl_species,
  file = "figures/species_plot.png", height = 7, width = 9)
ggsave(pl_species,
  file = "figures/species_plot.pdf", height = 7, width = 9)
```

## 4 Species abundances

Reviewers asked about relative abundance of species. Analysing the tree plot data here.

```
trees <- read.csv("data/tree_plots.csv")
head(trees)
```

```
##   site location group square canopy  nets  plots      species
## 1  S13      L1    G2     5    70% 1 Net  1 plot      Acacia caesia
## 2  S13      L1    G2    15    50%           Actinodaphne bourdillonii
## 3  S13      L1    G2     2    45%           Actinodaphne bourdillonii
## 4  S13      L1    G2    15    50%           Actinodaphne bourdillonii
## 5  S19      L1    G2     2    75% 1 Net  2 Plots Actinodaphne bourdillonii
## 6  S19      L1    G2     4    70%           Actinodaphne bourdillonii
##   girth
## 1   0.1
## 2   0.2
## 3   6.0
```

```
## 4    2.0
## 5    0.6
## 6    0.8
```

```
summary(trees)
```

```
##      site      location      group      square
## Length:12096 Length:12096 Length:12096 Min.   : 1.000
## Class :character Class :character Class :character 1st Qu.: 4.000
## Mode  :character Mode  :character Mode  :character Median : 8.000
##                                         Mean  : 8.098
##                                         3rd Qu.:12.000
##                                         Max.   :16.000
##      canopy      nets      plots      species
## Length:12096 Length:12096 Length:12096 Length:12096
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##      girth
## Min.   : 0.100
## 1st Qu.: 0.500
## Median : 1.300
## Mean   : 3.735
## 3rd Qu.: 3.000
## Max.   :315.000
```

```
trees <- mutate(trees, species = str_trim(species))
abund <- trees |> group_by(species) |>
  summarize(q95_size = quantile(girth, 0.95),
            max_size = max(girth),
            n = length(girth),
            n_large = sum(girth > 30),
            n_sapling = sum(girth < 1)) |>
# filter(q95_size > 10) |>
  arrange(desc(n))

abund_15 <- abund[1:15,]
write_csv(abund_15, "tables/s1_specieslist.csv")

## correlation with abundance
table(trees$group, trees$site, trees$location)
```

```
## , , = L1
##
##
##      S12 S13 S18 S19  S2  S3  S5  S8
## G2    0 2519 2643 1406 537 1545  0 1488
## G3    0   0   0   0   0   0   0   0
##
## , , = L2
##
```

```
##
##      S12 S13 S18 S19 S2 S3 S5 S8
## G2    0  0  0  0  0  0 964  0
## G3  994  0  0  0  0  0  0  0
```

```
trees2 <- trees |> group_by(site, location, species) |> count()

trees2 <- as_tibble(trees2) |>
  complete(nesting(site, location), species, fill = list(n = 0))

trees2 <- trees2|> left_join(site_dat)
```

```
## Joining with 'by = join_by(site, location)'
```

```
## examine relationships between most common species and frag area
## none appear to respond to fragment area
```

```
trees2 <- filter(trees2, species %in% unique(abund_15$species))

sp_abund_mods <- map(unique(trees2$species), \(x)
  glmmTMB(n ~ scale(fragment.size),
    family = nbinom2,
    data = filter(trees2, species == x)))

length(sp_abund_mods)
```

```
## [1] 15
```

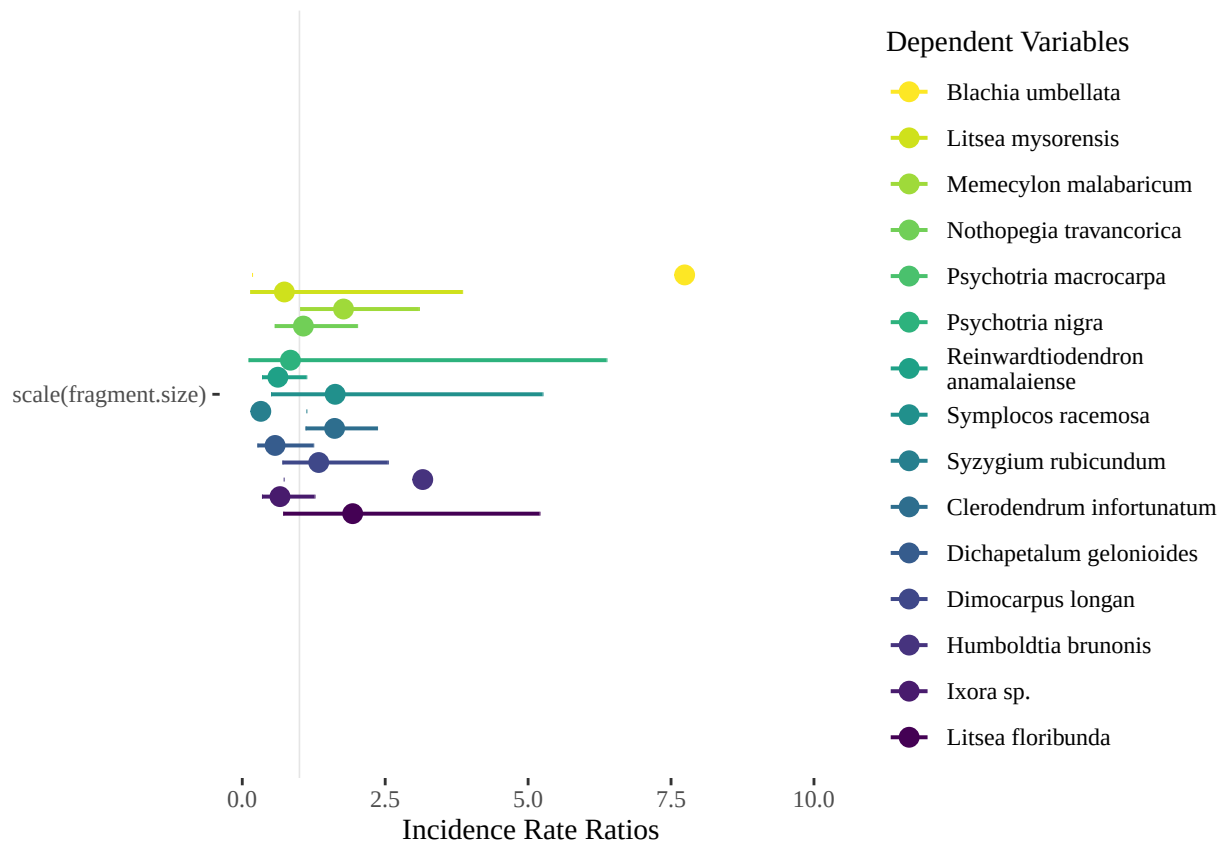
```
names(sp_abund_mods) <- unique(trees2$species)

plot_models(sp_abund_mods, m.labels=unique(trees2$species)) +
  ylim(0.1, 10) + scale_color_viridis_d()
```

```
## Warning in RColorBrewer::brewer.pal(n, pal): n too large, allowed maximum for palette Set1 is 9
## Returning the palette you asked for with that many colors
```

```
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
sp_abund_trend <- tibble(sp_bin = names(sp_abund_mods),
  bind_rows(map(sp_abund_mods, \(x) summary(x)$coef$cond[2,])))
sp_abund_trend
```

```
## # A tibble: 15 x 5
##   sp_bin                Estimate 'Std. Error' 'z value' 'Pr(>|z|)'
##   <chr>                  <dbl>         <dbl>    <dbl>    <dbl>
## 1 Blachia umbellata      2.05          1.94     1.05    0.293
## 2 Clerodendrum infortunatum -0.306        0.844    -0.363   0.717
## 3 Dichapetalum gelonioides  0.571        0.284     2.01    0.0443
## 4 Dimocarpus longan      0.0664       0.322     0.206   0.837
## 5 Humboldtia brunonis    -13.8        40.2     -0.343   0.732
## 6 Ixora sp.             -0.171       1.03     -0.166   0.868
## 7 Litsea floribunda      -0.473       0.299    -1.58    0.114
## 8 Litsea mysorensis      0.486       0.598     0.812   0.417
## 9 Memecylon malabaricum  -1.13       0.633    -1.78    0.0753
## 10 Nothopegia travancorica  0.480       0.193     2.49    0.0128
## 11 Psychotria macrocarpa   -0.555      0.394    -1.41    0.159
## 12 Psychotria nigra       0.291      0.329     0.885   0.376
## 13 Reinwardtiodendron anamalaiense 1.15       0.751     1.53    0.126
## 14 Symplocos racemosa     -0.416      0.332    -1.25    0.210
## 15 Syzygium rubicundum    0.658      0.505     1.30    0.193
```

```
## only 3 of 15 show a significant relationship.
```

## 5 Session Information

```
sessionInfo()
```

```
## R version 4.4.0 (2024-04-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 10 x64 (build 17763)
##
## Matrix products: default
##
## locale:
## [1] LC_COLLATE=English_United States.1252
## [2] LC_CTYPE=English_United States.1252
## [3] LC_MONETARY=English_United States.1252
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.1252
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] patchwork_1.2.0      sjPlot_2.8.16      ggdist_3.3.2
## [4] ggeffects_1.6.0.2    broom.mixed_0.2.9.5 DHARMA_0.4.6
## [7] glmmTMB_1.1.9        knitr_1.47          ggthemes_5.1.0
## [10] lubridate_1.9.3      forcats_1.0.0      stringr_1.5.1
## [13] dplyr_1.1.4          purrr_1.0.2         readr_2.1.5
## [16] tidyr_1.3.1          tibble_3.2.1        ggplot2_3.5.1
## [19] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gridExtra_2.3         inline_0.3.19       rlang_1.1.3
## [4] magrittr_2.0.3        frrrr_0.3.1         matrixStats_1.3.0
## [7] compiler_4.4.0        mgcv_1.9-1          loo_2.7.0
## [10] systemfonts_1.1.0     vctr_0.6.5          pkgconfig_2.0.3
## [13] crayon_1.5.2          fastmap_1.2.0       backports_1.5.0
## [16] labeling_0.4.3        effectsize_0.8.8    utf8_1.2.4
## [19] rmarkdown_2.27        tzdb_0.4.0          haven_2.5.4
## [22] nloptr_2.0.3          ragg_1.3.2          bit_4.0.5
## [25] xfun_0.44             jsonlite_1.8.8      highr_0.11
## [28] sjmisc_2.8.10         broom_1.0.6         parallel_4.4.0
## [31] R6_2.5.1              stringi_1.8.4       RColorBrewer_1.1-3
## [34] StanHeaders_2.32.9    parallelly_1.37.1   boot_1.3-30
## [37] numDeriv_2016.8-1.1  estimability_1.5.1  Rcpp_1.0.12
## [40] rstan_2.32.6          parameters_0.21.7    Matrix_1.7-0
## [43] splines_4.4.0         timechange_0.3.0    tidyselect_1.2.1
## [46] rstudioapi_0.16.0     yaml_2.3.8          TMB_1.9.11
## [49] codetools_0.2-20     sjlabelled_1.2.0    curl_5.2.1
## [52] listenv_0.9.1         pkgbuild_1.4.4      lattice_0.22-6
```

|         |                      |                    |                    |
|---------|----------------------|--------------------|--------------------|
| ## [55] | withr_3.0.0          | bayestestR_0.13.2  | coda_0.19-4.1      |
| ## [58] | evaluate_0.23        | future_1.33.2      | RcppParallel_5.1.7 |
| ## [61] | pillar_1.9.0         | stats4_4.4.0       | insight_0.20.5     |
| ## [64] | distributional_0.4.0 | generics_0.1.3     | vroom_1.6.5        |
| ## [67] | hms_1.1.3            | munsell_0.5.1      | scales_1.3.0       |
| ## [70] | minqa_1.2.7          | globals_0.16.3     | xtable_1.8-4       |
| ## [73] | glue_1.7.0           | emmeans_1.10.2     | tools_4.4.0        |
| ## [76] | lme4_1.1-35.3        | mvtnorm_1.2-5      | grid_4.4.0         |
| ## [79] | QuickJSR_1.2.0       | datawizard_0.10.0  | colorspace_2.1-0   |
| ## [82] | nlme_3.1-164         | performance_0.11.0 | beeswarm_0.4.0     |
| ## [85] | cli_3.6.2            | textshaping_0.4.0  | fansi_1.0.6        |
| ## [88] | viridisLite_0.4.2    | sjstats_0.19.0     | V8_5.0.0           |
| ## [91] | ggh4x_0.2.8          | gtable_0.3.5       | digest_0.6.35      |
| ## [94] | farver_2.1.2         | htmltools_0.5.8.1  | lifecycle_1.0.4    |
| ## [97] | bit64_4.0.5          | MASS_7.3-60.2      |                    |